

Executive Summary

Ragent is an open-source AI coding agent for the terminal, written entirely in Rust and distributed as a single statically-linked binary with zero external runtime dependencies. It orchestrates multiple LLM providers — Anthropic, OpenAI, GitHub Copilot, Ollama (local and cloud), and any OpenAI-compatible endpoint — behind a unified streaming interface, giving developers a powerful, provider-agnostic assistant that runs wherever a terminal does.

What It Does

Ragent bridges the gap between conversational AI and hands-on software engineering. An agent can read and write files, execute shell commands, search codebases, manage Git and GitHub workflows, query language servers, read and write office documents, and coordinate with other agents — all through a large built-in tool library organised across multiple categories. Every tool invocation passes through a multi-layered security and permission system that gives the user full control over what the agent can and cannot do.

How It Works

At its core, ragent follows a **session** → **agent** → **tool** loop. A session processor manages the conversation with the LLM provider, the agent system defines personality and capabilities via profiles, and the tool registry dispatches execution requests. An asynchronous event bus (built on tokio) connects all components, enabling real-time streaming of tokens, tool results, and status updates to both the TUI and the HTTP API.

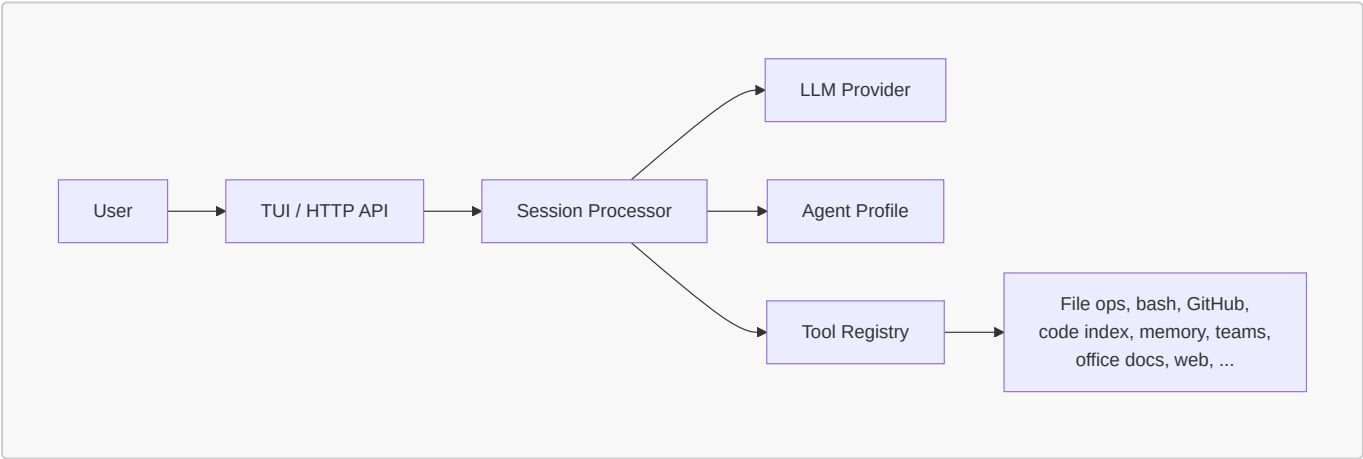


Figure 1: Core Execution Loop — High-level session/agent/tool flow

Key Capabilities

Capability	Summary
Multi-provider LLM	7 providers with automatic model discovery, health monitoring, streaming, vision, and reasoning levels
Terminal UI	Full-screen ratatui interface with streaming markdown, syntax highlighting, slash commands, and image support
HTTP Server	REST + SSE API (Axum) for headless operation and external integrations

Capability	Summary
Tool System	Broad tool coverage across file ops, shell, search, GitHub, GitLab, code index, memory, teams, sub-agents, office/PDF, web, and MCP
Code Intelligence	Tree-sitter parsing (15+ languages), Tantivy FTS, symbol/reference search, and code index queries
Persistent Memory	Three-tier system — file blocks, structured SQLite store, and optional embedding-based semantic search — with automatic extraction, decay, compaction, and a knowledge graph
Teams & Swarms	Multi-agent coordination with named teammates, shared task lists, mailbox messaging, and swarm decomposition for parallel work
Security	Permission rules (allow/deny/ask), 7-layer bash safety, file-path guards, secret redaction, resource limits, and YOLO mode for trusted environments
Skills	Loadable skill packs (bundled or custom YAML) that inject tools, prompts, and file context into agent sessions
Custom Agents	OASF-based agent profiles with configurable models, tools, permissions, and personality
Autopilot	Autonomous operation mode with configurable iteration limits and permission auto-approval

Who It's For

Ragent is designed for software developers and teams who want an AI assistant that lives in their terminal, respects their security boundaries, and learns from their workflow over time. It is equally suited to interactive pair-programming sessions and headless CI/CD integration via its HTTP API.

Technology

Aspect	Detail
Language	Rust (edition 2024)
Async runtime	tokio
TUI framework	ratatui + crossterm
HTTP framework	Axum
Database	SQLite (rusqlite, compiled-in)
Full-text search	Tantivy
Code parsing	tree-sitter (15+ grammars compiled-in)
Embeddings	ONNX Runtime (optional, all-MiniLM-L6-v2)
Binary size	Single static binary, ~50 MB release

Aspect	Detail
Platforms	Linux, macOS, Windows (cross-compiled)

Project Status

Ragent is in **alpha** (v0.1.0-alpha.82). The core architecture, tool system, TUI, HTTP server, memory system, teams, and security layer are functional and under active development. The specification below documents the current state of all subsystems.

Current Release Highlights:

- **Permission System Milestones Complete:**
 - Milestone 1: Core Permission System (7 tasks, 20 tests passing)
 - Milestone 2: Bash Security — 7 Layers (8 tasks, 27+ tests passing)
- Permission dialog countdown timer now redraws live in the TUI with a 120-second timeout and **EXPIRED** state
- Slash-command autocomplete now closes cleanly on **Esc** while preserving input and clamping the cursor safely
- Config parse errors now report the file path, line, column, problematic line, and a caret marker for faster recovery
- Codeindex tools are hardwired as always-allowed read-only tools and no longer trigger permission prompts
- Workspace crate reorganisation milestones extracted **ragent-types**, **ragent-config**, **ragent-storage**, and **ragent-llm**
- Broad tool coverage including comprehensive team coordination tools
- Native GitLab integration with issues, merge requests, and CI/CD pipeline management
- **Azure AI Foundry provider** — New **azure_foundry** provider for Microsoft Azure AI Foundry with OpenAI-compatible endpoints, dynamic model discovery, streaming, tool calling, vision, and reasoning levels
- **Azure endpoint logging** — Full endpoint URL displayed in TUI log panel for Azure AI Foundry requests
- **/config show slash command** — Displays current resolved configuration in the TUI
- **SPEC.md mermaid diagram fixes** — All 14 diagrams now pass syntax validation
- **gen-spec-pdf.sh script** — Pandoc + Chromium-based Markdown-to-PDF conversion for specifications
- **Startup ASCII art banner** — Application name rendered in ASCII art on TUI startup with compile timestamp
- **/codeindex lang filtering** — Optional language parameter for code index results (e.g., **/codeindex lang rust**)
- **Instruction file discovery logging** — Tracks which **AGENTS.md**-style files were found and where, with discovery summary logging

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Part I: Foundation & Basics

1. Overview

Ragent is an AI coding agent for the terminal, built in Rust. It provides multi-provider LLM orchestration, a built-in tool system, terminal UI, and client/server architecture — all compiled into a single statically-linked binary.

1.1 Key Characteristics

- **Multi-provider LLM support** — Anthropic, OpenAI, GitHub Copilot, Ollama, and Generic OpenAI-compatible APIs

- **Comprehensive tool system** — Extensive coverage across file operations, code analysis, GitHub/GitLab integration, web access, office documents, memory, teams, and more
- **Built-in TUI** — Full-screen ratatui interface with streaming chat, slash commands, and real-time updates
- **HTTP server** — REST + SSE API for external integrations
- **Zero external dependencies** — Self-contained binary with SQLite, Tantivy, and tree-sitter compiled in

2. Architecture

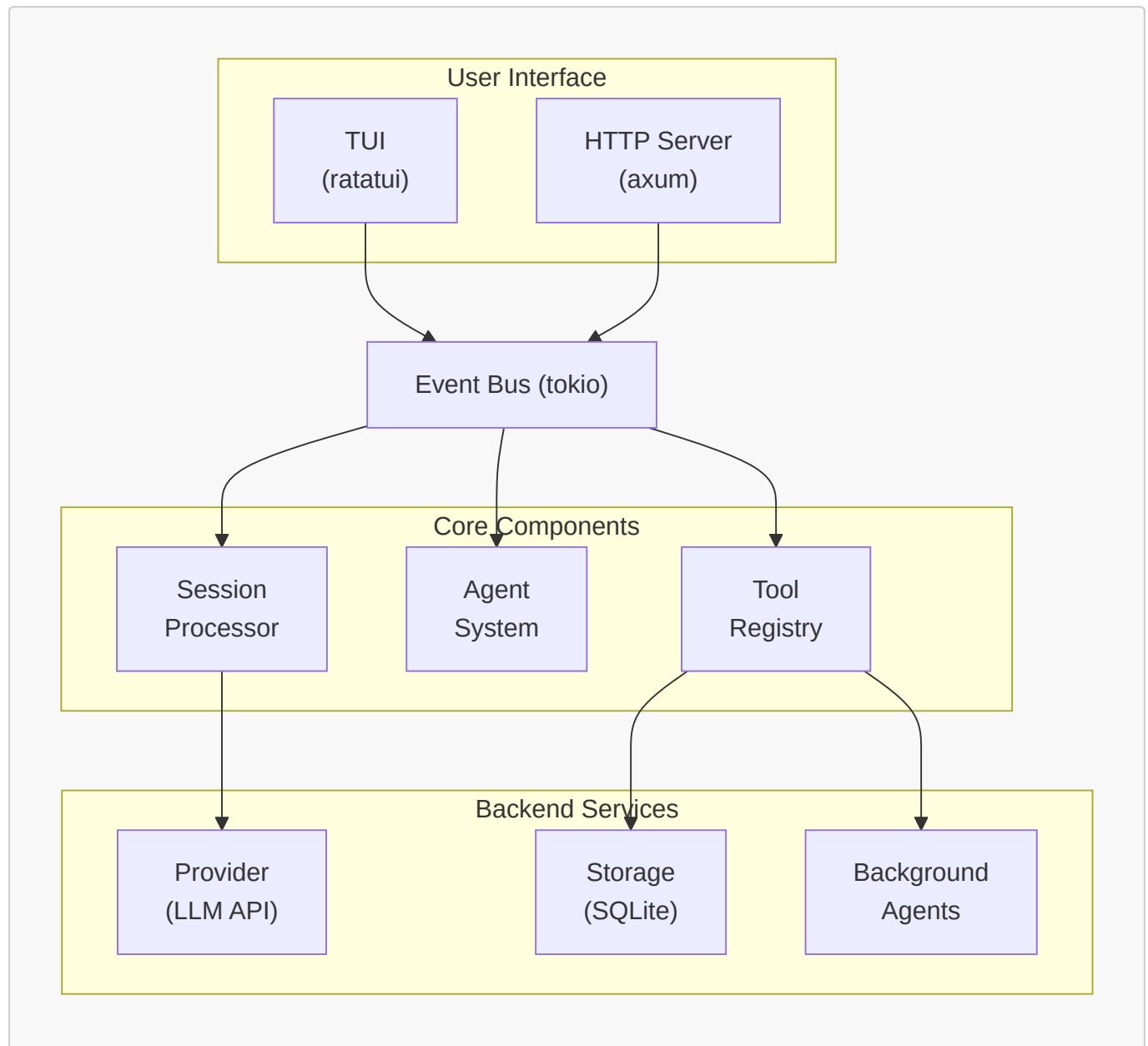


Figure 2: System Architecture — Full crate and component topology

2.3 Event Bus Flow

The event bus is a central tokio broadcast channel that connects all subsystems. Every component publishes events and subscribes to events it cares about.

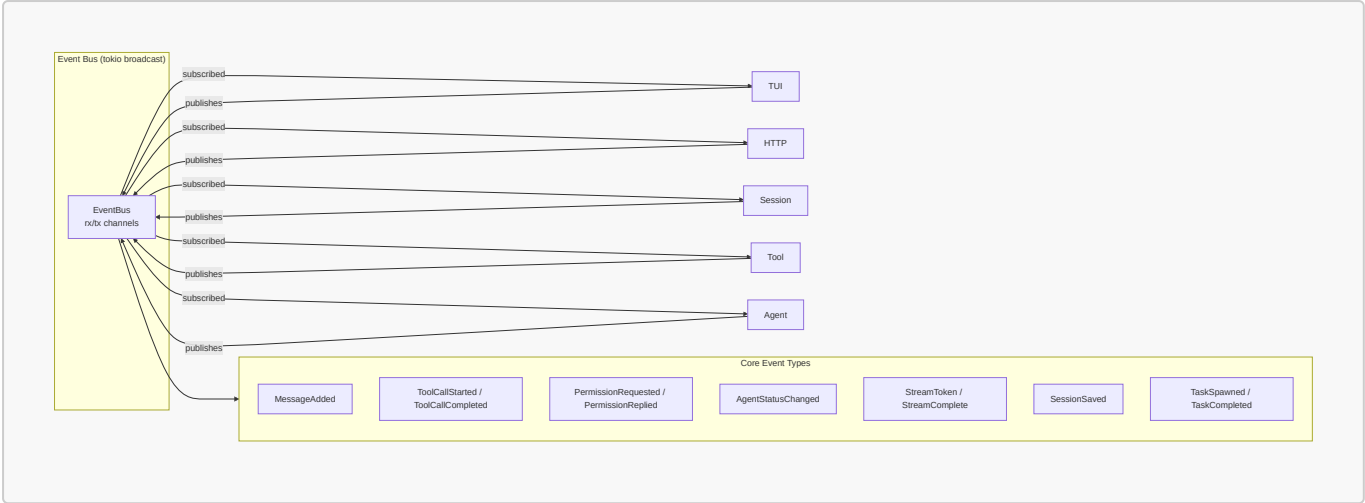


Figure 4: Event Bus Flow — Internal pub/sub message routing

Event Flow Example — Tool Execution:

- 1. **Session** sends tool call request to **Tool**
- 2. **Tool** publishes **ToolCallStarted** event
- 3. **TUI** receives event → updates log panel
- 4. **Tool** executes and publishes **ToolCallCompleted** with result
- 5. **Session** receives result → adds to conversation history
- 6. **Session** publishes **MessageAdded** → TUI updates chat panel

Crate	LOC %	Purpose
ragent-agent	34.61%	Agent/runtime layer: sessions, orchestration, MCP, memory, tool registry
ragent-codeindex	9.11%	Codebase indexing: tree-sitter parsing, SQLite store, Tantivy FTS, file watcher
ragent-config	1.29%	Configuration types, defaults, and parsing
ragent-llm	4.04%	Provider clients and model/provider registry
ragent-prompt_opt	0.40%	Prompt optimization transformations
ragent-server	2.47%	Axum HTTP routes and SSE streaming
ragent-storage	1.70%	SQLite storage, snapshots, and encrypted credential persistence
ragent-team	3.63%	Team runtime, team state, and team tools
ragent-tools-core	3.56%	Core shell/file/search tools
ragent-tools-extended	7.08%	Extended document/web/memory/codeindex tools
ragent-tools-vcs	2.08%	GitHub and GitLab tool surface
ragent-tui	20.92%	Ratatui terminal interface
ragent-types	1.21%	Shared IDs, events, messages, and sanitization primitives

Crate	LOC %	Purpose
ragent-specs	—	Spec lifecycle management: discovery, validation, status transitions, review, archival
ragent-bench	—	Benchmark runner shared between TUI and CLI

Percentages are based on a fresh count of current Rust `.rs` lines across workspace crates (167,466 total).

2.2 Crate Dependency Graph

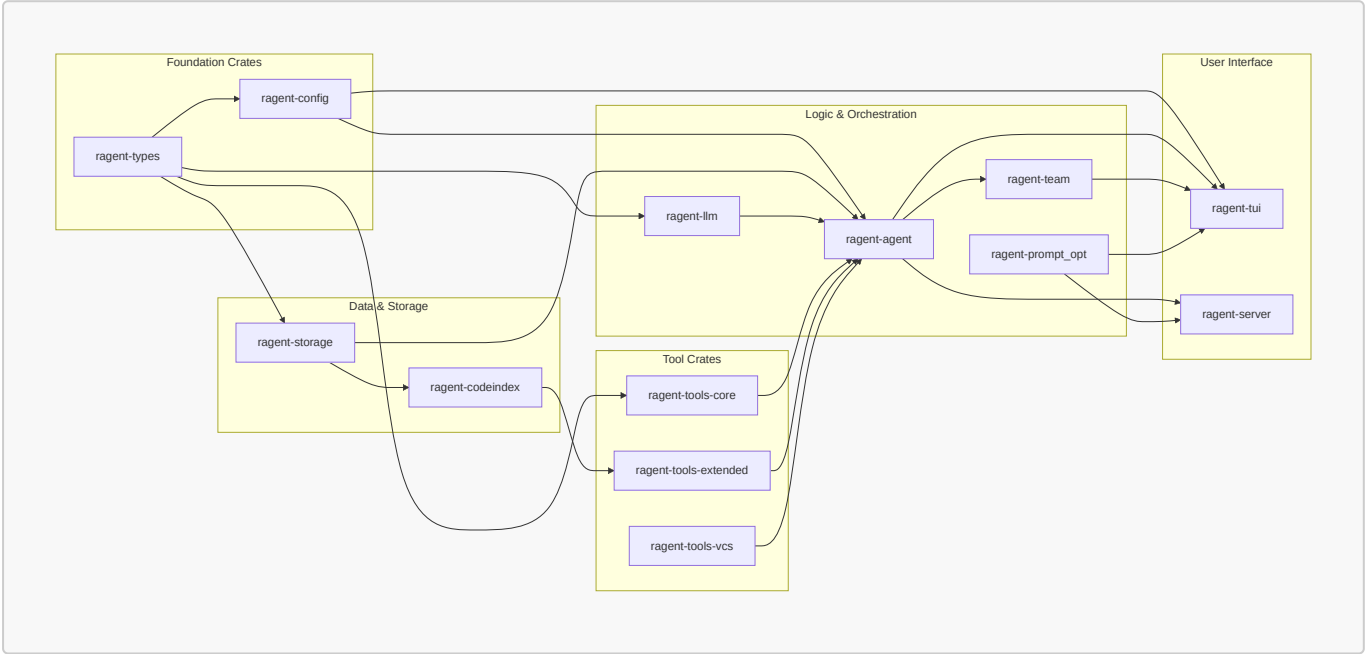


Figure 3: Crate Dependency Graph — Inter-crate dependency relationships

Dependency Rules:

- Foundation crates (`types`, `config`) have no internal dependencies
- `storage` depends only on `types`
- `llm` depends on `types` and `config`
- Tool crates depend on `types` (and `codeindex` for extended tools)
- `agent` is the integration layer — it depends on most other crates
- `team` depends on `agent` types
- `tui` and `server` are terminal layers that depend on `agent` and `team`
- Circular dependencies are prohibited; the graph is strictly acyclic

3. Core Features

3.1 LLM Providers

Supported Providers

Provider	ID	Authentication	Features
Anthropic	anthropic	ANTHROPIC_API_KEY	Streaming, tools, vision, reasoning
OpenAI	openai	OPENAI_API_KEY	Streaming, tools, vision
GitHub Copilot	copilot	Auto-discovered from VS Code	Streaming, tools, vision, reasoning levels
Ollama	ollama	No key required	Local models, streaming
Ollama Cloud	ollama_cloud	OLLAMA_API_KEY	Remote Ollama servers, dynamic model discovery, vision
Hugging Face	huggingface	HF_TOKEN	Streaming, tools, vision, dynamic model discovery
Generic OpenAI	generic_openai	GENERIC_OPENAI_API_KEY	Any OpenAI-compatible endpoint
Google Gemini	gemini	GEMINI_API_KEY	Streaming, tools, vision, reasoning
Azure AI Foundry	azure_foundry	AZURE_AI_FOUNDRY_API_KEY	OpenAI-compatible endpoints, dynamic model discovery, streaming, tools, vision, reasoning

Provider Features

- **Health indicators** — Real-time connectivity status (● green/✗ red/● yellow)
- **Model discovery** — Automatic model listing from provider APIs
- **Vision support** — Image attachments for supported models
- **Reasoning levels** — Copilot accepts reasoning_effort or reasoning_level with low/medium/high/none
- **Context window display** — Status bar shows context utilization percentage
- **Extended thinking** — Anthropic extended thinking/reasoning support
- **Usage tracking** — Token usage, quota percentage, and provider plan display where available
- **Dynamic model metadata** — Provider model pickers surface live-discovered context windows, capabilities, and Copilot premium request multipliers

Anthropic Models

Model	Context	Max Output	Capabilities
claude-sonnet-4-20250514	200,000	64,000	reasoning, streaming, vision, tool_use
claude-3-5-haiku-latest	200,000	8,192	streaming, vision, tool_use

OpenAI Models

Model	Context	Max Output	Capabilities
gpt-4o	128,000	16,384	streaming, vision, tool_use
gpt-4o-mini	128,000	16,384	streaming, vision, tool_use

Ollama Cloud Provider

The Ollama Cloud provider connects to remote Ollama servers using native `/api/chat` and `/api/tags` endpoints with Bearer token authentication.

Configuration:

- **Environment Variable:** `OLLAMA_API_KEY` — API key for authenticated Ollama Cloud instances
- **Default Endpoint:** `https://ollama.com`
- **Custom Endpoint:** Configurable via `base_url` in `ragent.json`

Features:

- **Dynamic Model Discovery** — Automatically fetches available models from `/api/tags` endpoint
- **Context Window Detection** — Queries `/api/show` to retrieve actual context length from model metadata
- **Vision Capability Detection** — Automatically detects vision support from model capabilities
- **Streaming Support** — Native SSE streaming via `/api/chat` endpoint
- **Tool Support** — Compatible with Ollama tool-calling format

Model Listing:

```
ragent models --provider ollama_cloud
```

Configuration Example (`ragent.json`):

```
{
  "provider": {
    "ollama_cloud": {
      "thinking": {
        "enabled": true,
        "level": "low"
      },
      "models": {
        "llama3.2": {
          "thinking": {
            "enabled": true,
            "level": "high"
          }
        }
      }
    }
  }
}
```

```
}  
}
```

Ollama (Local) Provider

The local Ollama provider connects to self-hosted Ollama instances (no authentication required for local servers).

Configuration:

- **Environment Variable:** `OLLAMA_HOST` (optional) — Custom server URL (default: `http://127.0.0.1:11434`)
- **No API Key Required** — Local Ollama servers run without authentication

Features:

- **Local Model Execution** — Run models on local hardware (CPU/GPU)
- **Dynamic Discovery** — Lists locally available models via `/api/tags` at runtime (placeholder defaults are only used as fallback metadata)
- **OpenAI-Compatible API** — Uses `/v1/chat/completions` endpoint
- **Streaming Support** — Full SSE streaming

Model Listing:

```
ragent models --provider ollama
```

Google Gemini Provider

The Google Gemini provider connects to Google's Gemini API for state-of-the-art multimodal models with extensive context windows.

Authentication: `GEMINI_API_KEY` environment variable

Default Models:

Model	Context	Cost (Input/Output)	Capabilities
<code>gemini-2.5-flash-preview-05-20</code>	1,048,576	\$0.15 / \$0.60	reasoning, streaming, vision, tool_use
<code>gemini-2.5-pro-preview-05-06</code>	1,048,576	\$1.25 / \$10.00	reasoning, streaming, vision, tool_use
<code>gemini-2.0-flash</code>	1,048,576	\$0.10 / \$0.40	streaming, vision, tool_use
<code>gemini-2.0-flash-lite</code>	1,048,576	\$0.075 / \$0.30	streaming, vision, tool_use
<code>gemini-1.5-flash</code>	1,048,576	\$0.075 / \$0.30	streaming, vision, tool_use

Model	Context	Cost (Input/Output)	Capabilities
<code>gemini-1.5-pro</code>	2,097,152	\$1.25 / \$5.00	reasoning, streaming, vision, tool_use

Features:

- **Streaming** — Real-time token-by-token response streaming
- **Tool Use** — Native function calling for all models
- **Vision** — Image understanding capabilities
- **Reasoning** — Available on Pro and Flash 2.5 models
- **Massive Context Windows** — Up to 2M tokens on 1.5 Pro

API Base: <https://generativelanguage.googleapis.com>

Hugging Face Provider

The HuggingFace provider connects to the HuggingFace Inference API, which exposes an OpenAI-compatible `/v1/chat/completions` endpoint. Supports both the free/Pro shared Inference API and dedicated Inference Endpoints.

Authentication:

- **Primary:** `HF_TOKEN` environment variable (standard HuggingFace token)
- **Legacy:** `HUGGING_FACE_HUB_TOKEN` (older HF token name)
- **Ragent convention:** `RAGENT_API_KEY_HUGGINGFACE` (auto-checked)

Default Models:

Model	Context	Capabilities
<code>meta-llama/Llama-3.1-8B-Instruct</code>	128,000	streaming, tool_use
<code>meta-llama/Llama-3.1-70B-Instruct</code>	128,000	streaming, tool_use
<code>mistralai/Mixtral-8x7B-Instruct-v0.1</code>	32,000	streaming, tool_use
<code>Qwen/Qwen2.5-72B-Instruct</code>	128,000	streaming, tool_use
<code>microsoft/Phi-3-mini-4k-instruct</code>	4,096	streaming

Features:

- **OpenAI-Compatible API** — Uses `/v1/chat/completions` endpoint (same as OpenAI)
- **Streaming Support** — Full SSE streaming with tool call deltas
- **Tool Use** — Function calling for models that support it (Llama 3.1+, Mixtral, Qwen)
- **Dynamic Model Discovery** — Queries HuggingFace Hub API for available text-generation models with warm inference endpoints (up to 50 models)
- **Model Loading Detection** — Detects 503 "model loading" responses with estimated wait time
- **Gated Model Handling** — Clear error messages for models requiring license acceptance
- **Rate Limit Tracking** — Parses `X-RateLimit-Limit/X-RateLimit-Remaining` headers

- **Tool Name Compatibility** — Internally prefixes tool names sent to the Hugging Face router to avoid streaming-mode name rejection, then maps responses back to canonical ragent tool names

Provider-Specific Options:

Option	Type	Default	Description
<code>wait_for_model</code>	bool	<code>true</code>	Send <code>x-wait-for-model: true</code> header to wait for cold models
<code>use_cache</code>	bool	<code>true</code>	Enable server-side response caching

Inference Endpoints:

For dedicated deployments, configure the custom endpoint URL:

```
{
  "provider": {
    "huggingface": {
      "api": {
        "base_url": "https://my-endpoint.endpoints.huggingface.cloud"
      }
    }
  }
}
```

Model Listing:

```
ragent models --provider huggingface
```

Azure AI Foundry Provider

The Azure AI Foundry provider connects to Microsoft Azure AI Foundry models via OpenAI-compatible endpoints with `api-key` header authentication. Supports dynamic model discovery, streaming chat completions, tool calling, vision, and reasoning levels.

Authentication:

- **Primary:** `AZURE_AI_FOUNDRY_API_KEY` environment variable
- **Base URL:** `AZURE_AI_FOUNDRY_BASE` environment variable (optional, for custom endpoints)

Configuration Example (`ragent.json`):

```
{
  "provider": {
    "azure_foundry": {
      "env": ["AZURE_AI_FOUNDRY_API_KEY"],
      "api": {
```

```
        "base_url": "https://your-endpoint.azure.com"
    },
    "thinking": {
        "enabled": true,
        "level": "low"
    }
}
}
```

Features:

- **OpenAI-Compatible API** — Uses `/v1/chat/completions` endpoint with `api-key` header
- **Dynamic Model Discovery** — Automatically fetches available models from Azure endpoint
- **Streaming Support** — Full SSE streaming
- **Tool Support** — Function calling compatible with OpenAI tool format
- **Vision Support** — Image understanding for vision-capable models
- **Reasoning Levels** — Supports `low`, `medium`, `high`, `none` reasoning effort (o1, o3-mini models)
- **Endpoint Logging** — Full resolved endpoint URL logged in TUI log panel for debugging

Model Listing:

```
ragent models --provider azure_foundry
```

3.2 Tool System

File Operations Tools (26)

Tool	Purpose
<code>read</code>	Read file contents with line range support
<code>write</code>	Create new files
<code>edit</code>	Replace text in existing files
<code>create</code>	Create new file (alternative to write)
<code>rm</code>	Delete single files
<code>move_file</code>	Move/rename files and directories
<code>copy_file</code>	Copy files to new location
<code>mkdir</code>	Create directories (<code>mkdir -p</code>)
<code>append_file</code>	Append text to end of file
<code>file_info</code>	Get metadata (size, mtime, type)
<code>diff_files</code>	Compare two files

Tool	Purpose
<code>glob</code>	Find files matching glob patterns
<code>list</code>	List directory contents
<code>multiedit</code>	Atomic multi-file edits
<code>patch</code>	Apply unified diff patches
<code>file_ops_tool</code>	Combined file operations

File Operation Aliases

The following are aliases for commonly requested operations:

Alias	Maps To
<code>read</code>	<code>read</code>
<code>list_files</code> , <code>list_directory</code>	<code>list</code>
<code>find_files</code>	<code>glob</code>
<code>update_file</code>	<code>edit</code>
<code>file_search</code>	<code>grep</code>

Execution Tools (10)

Tool	Purpose
<code>bash</code>	Execute shell commands with security restrictions
<code>bash_reset</code>	Reset bash shell state
<code>execute_python</code>	Run Python code snippets
<code>run_code</code> / <code>execute_code</code> / <code>execute_bash</code> / <code>run_shell_command</code> / <code>run_terminal_cmd</code>	Aliases for bash/code execution

Interactive Tools (3)

Tool	Purpose
<code>question</code> / <code>ask_user</code>	Interactive user prompts
<code>think</code>	Record reasoning notes (no-op)
<code>todo_read</code>	Read TODO items
<code>todo_write</code>	Manage TODO items

Utility Tools (3)

Tool	Purpose
calculator	Evaluate mathematical expressions
get_env	Read environment variables

3.2.1 Tool System Categories Summary

Category	Count	Description
File Operations	26	read, write, edit, create, rm, move, copy, mkdir, append, diff, multiedit, patch, etc.
Execution	10	bash, bash_reset, execute_python, aliases
Search	4	grep and aliases
Web	3	webfetch, websearch, http_request
Office	6	office_read/write/info, libre_read/write/info
PDF	2	pdf_read, pdf_write
Code Index	6	codeindex_search, symbols, references, dependencies, status, reindex
GitHub	10	Issues and PR management
GitLab	19	Issues, merge requests, pipelines, and jobs
Memory	12	memory_read/write/replace/store/recall/forget/search/migrate
Team	21	Team lifecycle, tasks, messaging, coordination
Sub-agent	5	new_task, cancel_task, list_tasks, wait_tasks, task_complete
Plan	2	plan_enter, plan_exit
MCP	1	mcp_tool (McpToolWrapper)
Interactive	4	question, think, todo_read/write
Utility	3	calculator, get_env

Team Tools (21)

Tool	Purpose
team_create	Create new team
team_spawn	Spawn teammate agent
team_cleanup	Cleanup team resources
team_status	Get team status
team_idle	Signal idle state
team_task_create	Create team task

Tool	Purpose
<code>team_task_claim</code>	Claim task to work on
<code>team_task_complete</code>	Mark task complete
<code>team_task_list</code>	List team tasks
<code>team_assign_task</code>	Assign task to specific teammate
<code>team_message</code>	Send message to team member
<code>team_broadcast</code>	Broadcast to all teammates
<code>team_read_messages</code>	Read mailbox messages
<code>team_shutdown_teammate</code>	Request teammate shutdown
<code>team_shutdown_ack</code>	Acknowledge shutdown request
<code>team_submit_plan</code>	Submit plan for approval
<code>team_approve_plan</code>	Approve teammate plan
<code>team_wait</code>	Wait for teammates to complete
<code>team_memory_read</code>	Read team memory
<code>team_memory_write</code>	Write to team memory

3.3 Agent System

Built-in Agents

Agent	Purpose	Tool Groups
<code>general</code>	General-purpose assistant	All tools
<code>coder</code>	Code-focused tasks	File, bash, search
<code>task</code>	Task execution	File, bash
<code>architect</code>	Design and planning	All tools
<code>ask</code>	Question answering	Read-only tools
<code>debug</code>	Debugging assistance	File, bash, search
<code>code-review</code>	Code review	Read, diff, github
<code>orchestrator</code>	Multi-agent coordination	All tools

Agent Features

- **Custom agents** — User-defined agents via JSON (OASF format) or Markdown profiles
- **Template variables** — Dynamic injection of context (`{{WORKING_DIR}}`, `{{FILE_TREE}}`, `{{AGENTS_MD}}`, `{{GIT_STATUS}}`, `{{README}}`)

- **Permission rules** — Per-agent access control for file paths and commands
- **Memory scoping** — Project-level and user-level memory for agents

3.3.1 GitHub Integration Tools

agent provides native GitHub issue and pull request tools that auto-detect the repository owner and name from the local git remote configuration.

Issue Tools

Tool	Description	Parameters
<code>github_issues_list</code>	List issues with filtering	<code>state</code> (open/closed/all), <code>labels</code> , <code>limit</code>
<code>github_issues_get</code>	Get issue details	<code>number</code>
<code>github_issues_create</code>	Create a new issue	<code>title</code> , <code>body</code> , <code>labels</code> , <code>assignees</code>
<code>github_issues_comment</code>	Add comment to an issue	<code>number</code> , <code>body</code>
<code>github_issues_close</code>	Close an issue	<code>number</code> , <code>comment</code> (optional)

Pull Request Tools

Tool	Description	Parameters
<code>github_pr_list</code>	List pull requests	<code>state</code> , <code>base</code> , <code>limit</code>
<code>github_pr_get</code>	Get PR details and diff	<code>number</code>
<code>github_pr_create</code>	Create a new pull request	<code>title</code> , <code>body</code> , <code>base</code> , <code>head</code> , <code>draft</code>
<code>github_pr_merge</code>	Merge a pull request	<code>number</code> , <code>method</code> (merge/squash/rebase)
<code>github_pr_review</code>	Submit a PR review	<code>number</code> , <code>event</code> (approve/comment/request_changes), <code>body</code>

Auto-Detection

Owner and repository are automatically detected from the git remote:

```
git remote get-url origin
→ https://github.com/owner/repo.git → owner="owner", repo="repo"
→ git@github.com:owner/repo.git    → owner="owner", repo="repo"
```

Falls back to explicit `--owner` and `--repo` parameters if detection fails.

3.5 Session & Tool Execution Flow

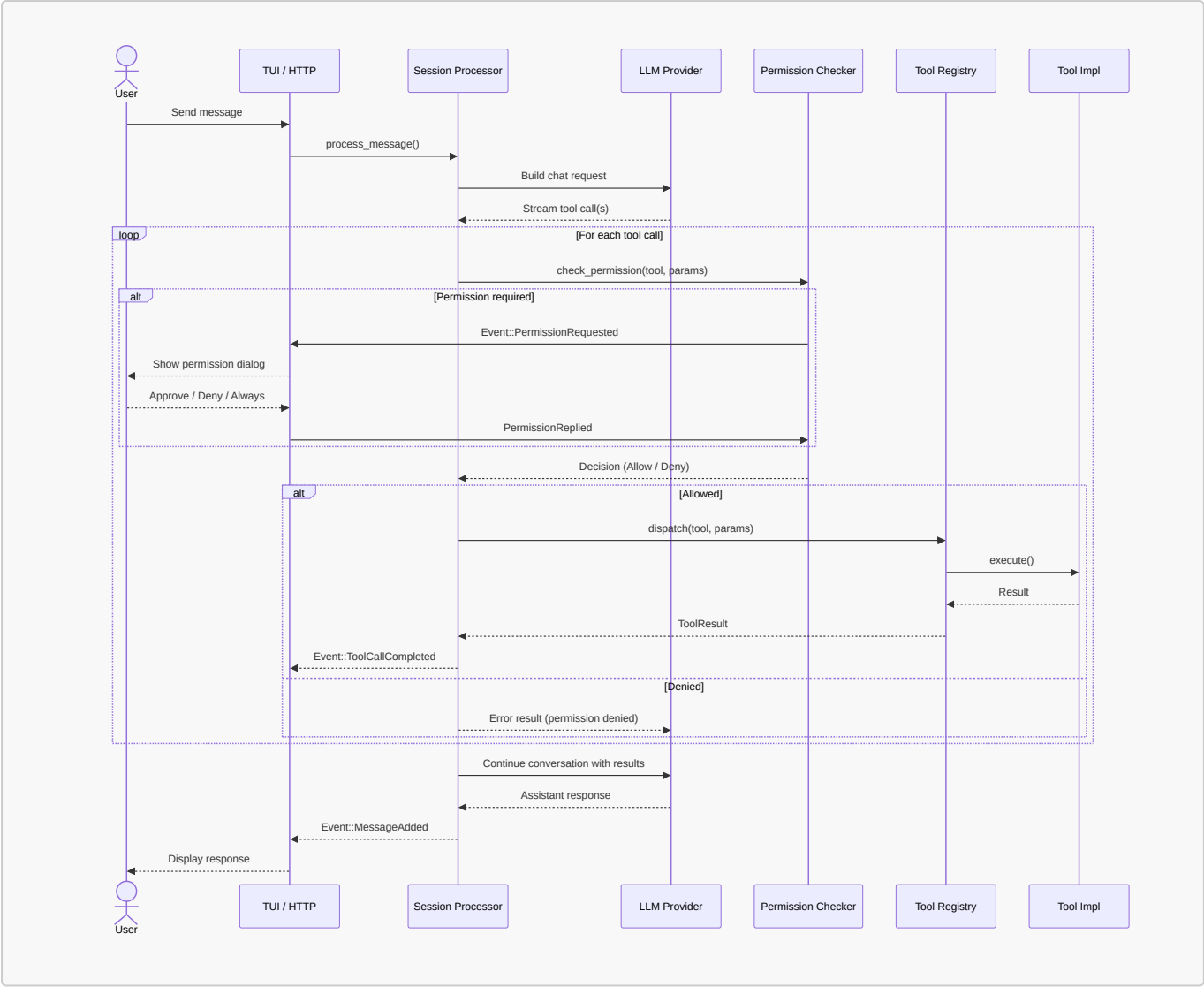
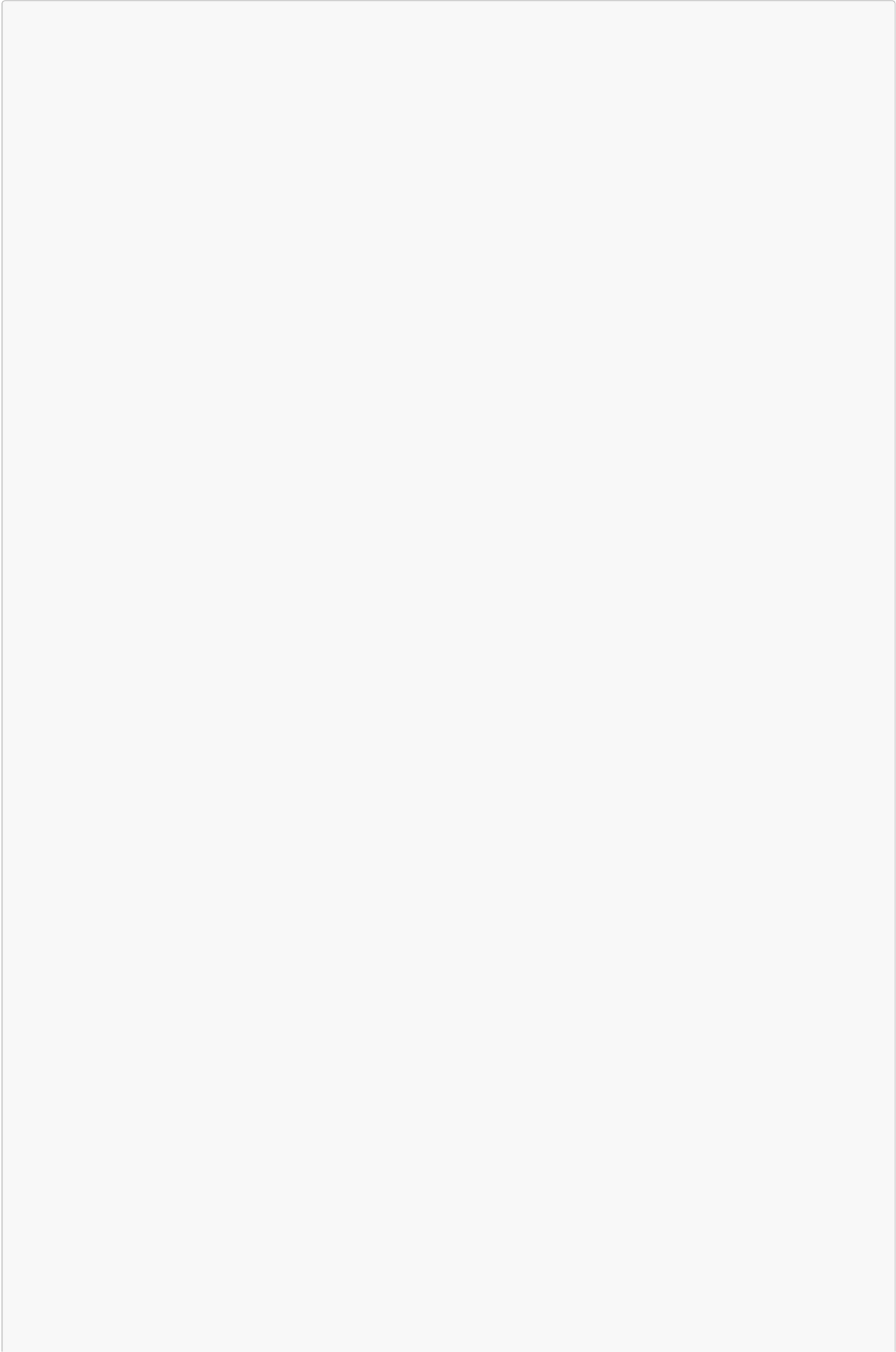
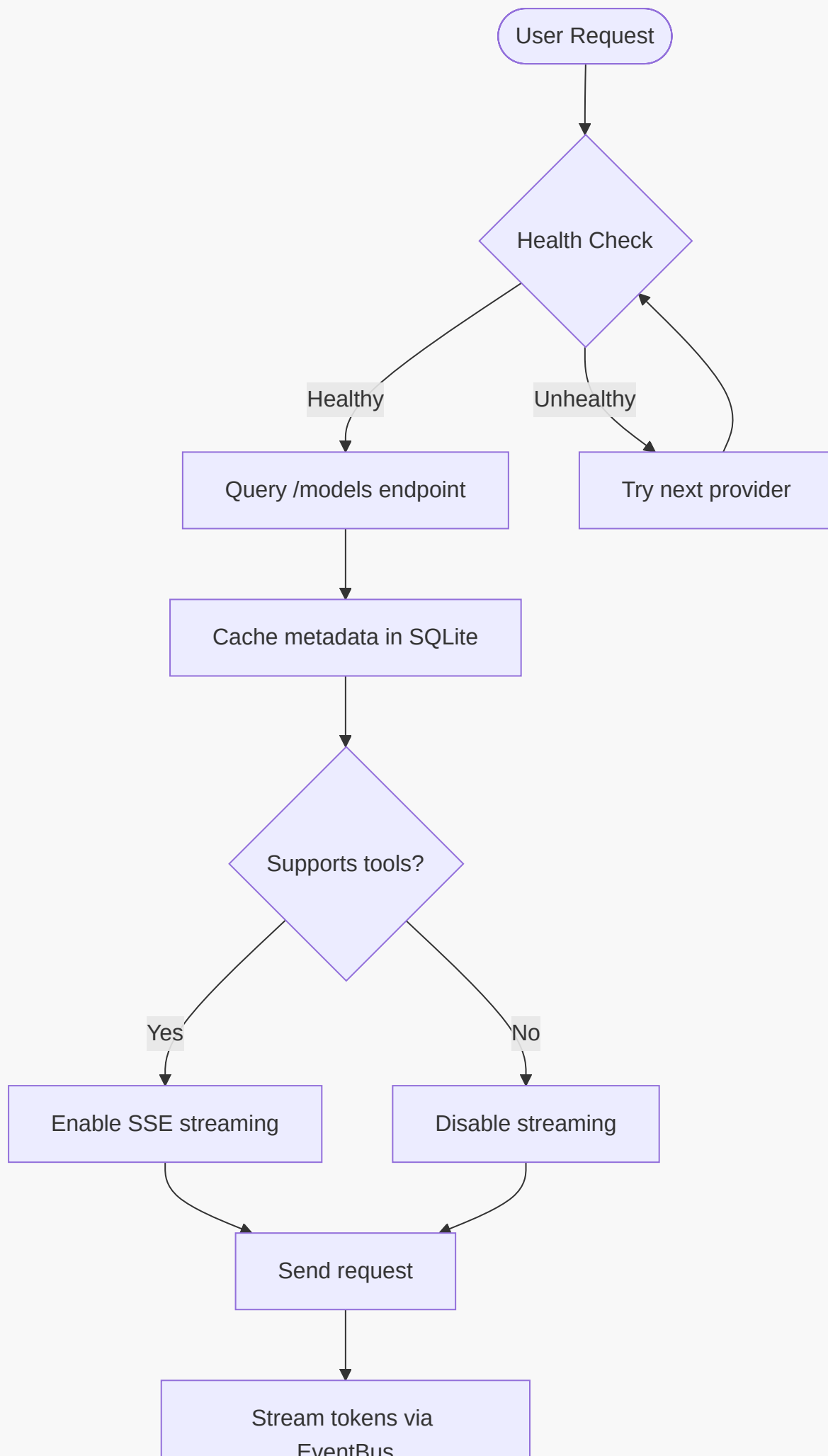


Figure 5: Session & Tool Execution Flow — LLM call → permission → tool dispatch loop

3.6 Provider Selection Flow





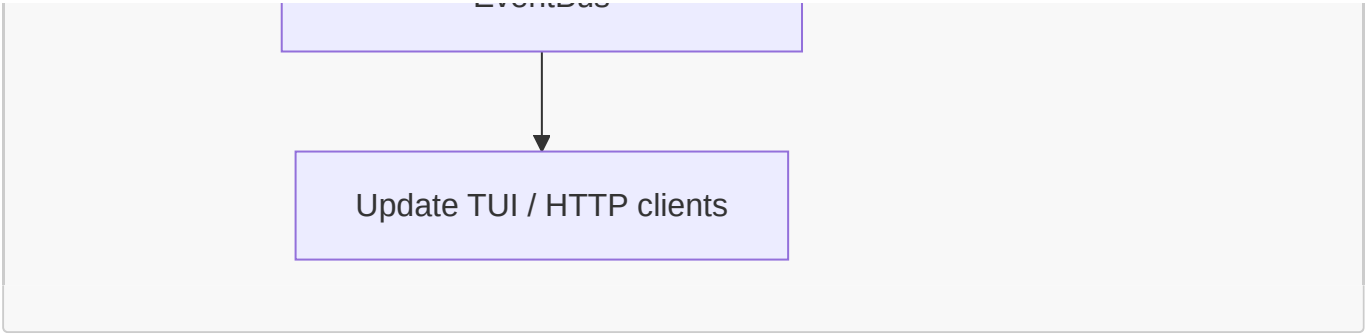


Figure 6: Provider Selection Flow — Multi-provider routing and health checks

3.7 Agent Execution Loop Phases

Each turn of an agent session follows a fixed pipeline of phases. The loop repeats until the LLM returns a final assistant message (no further tool calls), the user interrupts the session, or a safety limit (`max_steps`, token budget, or timeout) is reached.

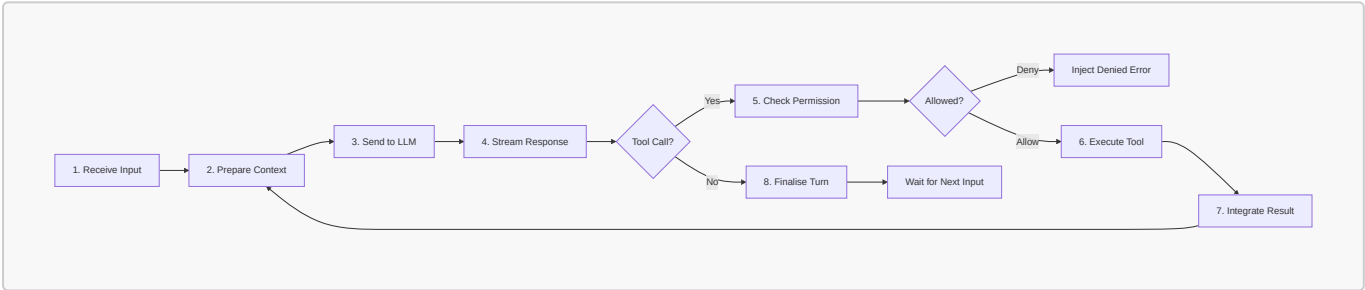


Figure 16: Agent Execution Loop Phases — One complete turn from input to response

Phase Descriptions

Phase	What Happens	Key Components
1. Receive Input	User message arrives via TUI (Enter) or HTTP POST. The session ID is resolved and the message is appended to the conversation history.	Session Processor, EventBus
2. Prepare Context	Build the chat request payload: system prompt (agent profile + AGENTS.md + injected variables), conversation history, available tool schemas, and any compaction/summarisation if near the context-window limit.	Agent Profile, Tool Registry, Context Compaction
3. Send to LLM	The configured provider client serialises the request (OpenAI, Anthropic, Gemini, etc. format), adds auth headers, and issues the HTTP request.	Provider Client, HTTP Client
4. Stream Response	Tokens arrive via SSE. The session processor forwards them to the EventBus as StreamToken events. If a tool call is emitted, streaming pauses and the loop transitions to Phase 5.	EventBus, SSE Stream

Phase	What Happens	Key Components
5. Check Permission	The permission checker evaluates the tool call against rules (hardwired → config → agent-specific → YOLO). Result can be Allow , Deny , or Ask (prompt user via TUI modal).	PermissionChecker, Permission Rules
6. Execute Tool	The tool registry dispatches the call to the tool implementation. The tool runs (file op, bash, web fetch, code index query, etc.) and returns a structured result.	Tool Registry, Tool Impl
7. Integrate Result	The tool result (or permission-denied error) is injected into the conversation history as a tool message. Control returns to Phase 2 for the next LLM call.	Session Processor
8. Finalise Turn	When the LLM produces a plain-text response with no tool calls, the assistant message is persisted, token usage is recorded, and the TUI displays the final output.	Storage, EventBus

Safety Limits

The loop is bounded by configurable guards:

Limit	Default	Behaviour When Hit
max_steps	500	Halt and ask user whether to continue
Token budget	Provider/model specific	Pause and warn; user can approve continuation
Timeout	Per-request configurable	Abort the running request and surface error
Context window	Model-specific	Trigger automatic compaction before sending

Streaming Semantics

Responses are streamed token-by-token so the user sees progress in real time:

- 1. **StreamToken** events fire for every chunk.
- 2. If a tool call is detected mid-stream, the UI shows a collapsible tool-call summary.
- 3. Tool results are streamed back as they complete (for async/multi-tool scenarios).
- 4. **StreamComplete** signals the end of the turn.

- **Persistent storage** — SQLite-backed conversation history
- **Session commands** — **agent session list**, **resume**, **export**, **import**
- **Step numbering** — Session-prefixed step numbers (**[sid:step]**) for traceability
- **Context compaction** — Automatic pre-send context management near window limits

4. Terminal User Interface (TUI)

4.1 TUI Windows and Overlay Panels

The ragent TUI is built on a multi-layer architecture with a main chat screen, modal overlays, popup windows, and sidebar panels. Each window serves a specific purpose in the user workflow.

4.1.1 Main Screen (Chat)

The primary interface where all conversation happens.

Component	Description
Status Bar (Line 1)	Shows session ID, agent name, working directory, git branch, and current status message
Status Bar (Line 2)	Displays provider/model, quota or token usage, context utilization, active tasks, and service indicators such as code index
Messages Panel	Scrollable conversation history with syntax highlighting and formatted tool calls
Input Area	Multi-line text input with autocomplete support for slash commands and file references
Log Panel	Toggleable panel showing step-numbered tool calls with pretty-printed JSON
Active Agents Subpanel	Sidebar showing running background agents with progress indicators
Teams Subpanel	Sidebar displaying team members, their status, and message counts

Access: This is the default screen when ragent starts (after initial setup).

4.1.2 Provider Setup Dialog (Modal)

Multi-step wizard for configuring LLM providers.

Step	Description
Select Provider	Choose from Anthropic, OpenAI, GitHub Copilot, Ollama, Ollama Cloud, or Generic OpenAI
Enter API Key	Secure input with masked characters and endpoint URL entry for Generic OpenAI
Device Flow	GitHub Copilot OAuth flow with user code and verification URL
Select Model	Browse available models with metadata (context window, cost, capabilities, and Copilot premium request multiplier where available)
Select Agent	Choose default agent personality
Reset Provider	Remove stored credentials for a provider
Done	Confirmation screen showing configured provider and model

Access: `/provider` command, or auto-triggered at first startup

4.1.4 Agents Popup Window

A floating popup window showing active background agents and their status.

Purpose: Monitor and switch between multiple concurrent agent sessions.

Features:

- List of active agents with session IDs
- Agent status indicators (running, idle, error)
- Message count per agent
- Click to focus specific agent session
- Close button to dismiss

Access: Click "Agents" button or press **a**

4.1.5 Teams Popup Window

A floating popup for team coordination when managing multiple teammates.

Purpose: Coordinate work across a team of specialized agents.

Features:

- Team member list with status
- Message counts (sent/received per teammate)
- Focus indicator for active teammate
- Task assignment interface
- Broadcast messaging capability

Access: Click "Teams" button or press **F10**

4.1.6 Slash Command Autocomplete Menu

An inline popup menu that appears when typing **/** in the input area.

Purpose: Quick discovery and selection of slash commands.

Features:

- Real-time filtering as you type
- Command descriptions
- Skill vs. builtin command indicators
- Keyboard navigation (**↑/↓**) and Enter to select
- **Esc** closes the menu while preserving the partially typed input and keeping the cursor within valid bounds

Access: Type **/** in input area

4.1.7 File Reference Autocomplete Menu (@ Menu)

An inline popup for selecting files when using @ references.

Purpose: Quickly reference files in the conversation.

Features:

- Fuzzy file search across project
- Directory navigation mode
- Hidden file toggle
- Recently used files prioritized
- Preview of selected file

Access: Type @ in input area, optionally followed by partial filename

4.1.8 History Picker Overlay

A scrollable overlay for browsing and reusing previous inputs.

Purpose: Quickly recall and resend previous prompts.

Features:

- Chronological list of previous inputs
- Search/filter capability
- Enter to insert, Esc to cancel
- Persistent across sessions (stored in SQLite)

Access: /history command or Up arrow with empty input

4.1.9 Permission Dialog (Modal)

Centered modal for approving or denying permission requests.

Purpose: Security gate for file writes, shell commands, and external access.

Features:

- Permission type indicator (file:write, bash:execute, etc.)
- Target path or command preview
- One-time (y/n) or always allow options
- Question mode with text input for user prompts

Access: Auto-triggered when tool requires permission

4.1.10 Context Menu (Right-Click)

A small popup menu for text operations.

Purpose: Standard text editing operations in any pane.

Features:

- Cut selected text
- Copy to clipboard
- Paste from clipboard
- Context-aware (disabled when no selection)

Access: Right-click in any pane

4.1.11 MCP Discovery Dialog (Overlay)

An overlay for discovering Model Context Protocol servers.

Purpose: Extend tool capabilities via MCP servers.

Features:

- Numbered list of discovered MCP servers
- Server metadata display
- Number input to connect
- Connection feedback

Access: `/mcp discover` command

4.1.12 Output View Overlay

A scrollable panel for viewing raw agent or team member output.

Purpose: Inspect unformatted output from specific agents or team members.

Features:

- Session output viewer
- Team member output viewer
- Scrollable content
- Syntax highlighting for code

Access: Auto-triggered for certain tool outputs or team member responses

4.1.15 Memory Browser Overlay

A full-panel overlay for browsing memory blocks.

Purpose: View and manage persistent memory across sessions.

Features:

- List of global and project memory blocks

- Size indicators (with warnings for blocks near limit)
- Expand/collapse to view full content
- Keyboard navigation (j/k, Enter, Esc)
- Search and filter capabilities

Access: `/memory` command

4.1.16 Plan Approval Dialog (Modal)

A centered dialog for approving or rejecting plans from the plan agent.

Purpose: Human-in-the-loop approval for plan agent proposals.

Features:

- Plan text display with scrollable content
- Approve/Reject buttons
- Cursor navigation between options
- On approve: switches to plan agent and executes
- On reject: returns to previous agent

Access: Auto-triggered when plan agent submits a plan

4.1.18 Force-Cleanup Confirmation Modal

A confirmation dialog for destructive team cleanup operations.

Purpose: Prevent accidental data loss when force-cleaning team resources.

Features:

- Warning message with team name
- List of active members that will be affected
- Explicit confirmation required
- Cancel option

Access: Triggered by `/team cleanup` when team has active members

4.1.19 Keybindings Help Panel (Overlay)

A scrollable help panel showing all keyboard shortcuts.

Purpose: Quick reference for TUI controls.

Features:

- Categorized keybindings
- Context-aware help (shows relevant shortcuts)
- Search within help

- Scroll with arrow keys

Access: `?` key when input is empty, or `/help` command

4.1.20 Session/Message Widget Overlays

Various inline widgets rendered within the message panel.

Widget	Purpose
MessageWidget	Renders individual chat messages with markdown formatting, syntax highlighting, and inline tool call summaries
Tool Result Summaries	Collapsible sections showing tool execution results
File Diff Widgets	Side-by-side or inline diffs for file edits
Image Widgets	Renders attached images with dimensions and preview

4.1.21 Window State Summary

State Field	Window	Access
provider_setup	Provider Setup Dialog	<code>/provider</code> , startup
show_agents_window	Agents Popup	Click "Agents" button, <code>a</code> key
show_teams_window	Teams Popup	Click "Teams" button, <code>F10</code> key
slash_menu	Slash Command Menu	Type <code>/</code> ; <code>Esc</code> closes without clearing the partially typed command
file_menu	File Reference Menu	Type <code>@</code>
history_picker	History Picker	<code>/history</code> , Up arrow
permission_queue	Permission Dialog	Auto (tool permission)
context_menu	Right-Click Menu	Right-click
mcp_discover	MCP Discovery	<code>/mcp discover</code>
output_view	Output View	Auto (tool output)
memory_browser	Memory Browser	<code>/memory</code>
plan_approval_pending	Plan Approval	Auto (plan submission)
pending_forcecleanup	Force-Cleanup Modal	<code>/team cleanup</code> (with active)

State Field	Window	Access
show_shortcuts	Keybindings Help	? (empty input), /help

4.2 Slash Commands

Command	Purpose
Core	
/about	Show application info, version, and authors
/help	Show available slash commands
/quit, /exit	Exit ragent
Session & Agent	
/agent <name>	Switch to specific agent
/agents	List all agents (built-in and custom)
/clear	Clear conversation history
/compact	Summarize and compact conversation history
/resume	Resume agent from halted state
/system <prompt>	Override agent system prompt
Provider & Model	
/model	Switch active model on current provider
/thinking auto off low medium high	Switch the active reasoning level for the selected model
/provider	Change LLM provider
/provider_reset	Reset provider and remove stored credentials
/llmstats	Show LLM response time and token throughput
/cost	Show token usage and estimated cost
Context & Config	
/context refresh	Clear cached file tree, git status, README
/browse_refresh	Refresh @ file-picker project index
/reload [all config mcp skills agents]	Reload customizations
/init	Analyze project and write to PROJECT_ANALYSIS.md
Tasks	

Command	Purpose
<code>/tasks</code>	List active background tasks
<code>/cancel <id></code>	Cancel a background task
<code>/bench list</code>	List supported benchmark suites and profiles
<code>/bench init <suite-or-all-or-full></code>	Initialize benchmark data under <code>benches/data/<suite></code> in sample or full mode
<code>/bench show</code>	Show benchmark defaults and the selected model
<code>/bench run <target></code>	Start a background benchmark run for a suite, profile, or <code>all</code> target and write workbook results
<code>/bench status</code>	Show active or last benchmark run status
<code>/bench open last</code>	Show the latest benchmark workbook path(s) and summary
<code>/bench cancel</code>	Cancel the active benchmark run
<code>/abort</code>	Abort current running agent
Tools	
<code>/tools</code>	List available tools with parameters
<code>/bash allow <cmd></code>	Add command to bash allowlist
<code>/bash deny <cmd></code>	Add command to bash denylist
<code>/bash reset</code>	Reset bash shell state
Code Index	
<code>/codeindex on off</code>	Toggle code indexing
<code>/codeindex reindex</code>	Force full re-index
<code>/codeindex status</code>	Show index status
Spec Management	
<code>/spec create <id> <feature></code>	Generate SPEC.md + PLAN.md via explore agent
<code>/spec list [status]</code>	List specs with optional status filter
<code>/spec search <query></code>	Full-text search across SPEC.md, PLAN.md, and REVIEW.md
<code>/spec validate [spec-id]</code>	Validate EARS compliance; all specs if no ID
<code>/spec status <id> <status></code>	Transition a spec to a new lifecycle status
<code>/spec task <id></code>	List tasks for a spec

Command	Purpose
<code>/spec activate <id></code>	Activate a spec for context injection into agent prompts
<code>/spec deactivate</code>	Deactivate the active spec
<code>/spec coverage <id></code>	Show requirement coverage report with linked tasks
Memory	
<code>/memory</code>	Open memory browser
Team	
<code>/team create <name></code>	Create new team
<code>/team open <name></code>	Open existing team
<code>/team close</code>	Close team session
<code>/team delete <name></code>	Delete team
<code>/team clear</code>	Clear team state
<code>/team tasks</code>	Show team tasks table
<code>/team status</code>	Show team status
<code>/team message <to> <content></code>	Send message to teammate
<code>/team broadcast <content></code>	Broadcast to all teammates
<code>/team spawn <agent></code>	Spawn teammate agent
<code>/team cleanup</code>	Cleanup team resources
MCP	
<code>/mcp discover</code>	Discover MCP servers
<code>/mcp list</code>	List connected MCP servers
<code>/mcp call <server> <tool></code>	Call MCP tool
Optimization	
<code>/opt <method> <prompt></code>	Optimize prompt
<code>/opt help</code>	Show optimization methods
Swarm & Autopilot	
<code>/swarm <prompt></code>	Auto-decompose goal into parallel subtasks
<code>/swarm status</code>	Check swarm execution status
<code>/autopilot on [--max-tokens N] [--max-time N]</code>	Enable autonomous operation

Command	Purpose
<code>/autopilot off</code>	Disable autonomous operation
<code>/autopilot status</code>	Show autopilot status
<code>/yolo</code>	Toggle YOLO mode (bypass all restrictions)
Agent Modes & Planning	
<code>/mode <role></code>	Set agent role: architect, coder, reviewer, debugger, tester, off
<code>/plan <description></code>	Delegate planning to the plan agent
GitHub Integration	
<code>/github login</code>	Authenticate with GitHub
<code>/github logout</code>	Remove GitHub credentials
<code>/github status</code>	Show GitHub connection status
GitLab Integration	
<code>/gitlab setup</code>	Configure GitLab connection (instance URL + PAT)
<code>/gitlab logout</code>	Remove GitLab credentials
<code>/gitlab status</code>	Show GitLab connection status
Todos	
<code>/todos</code>	Show TODO items
Skills	
<code>/skills</code>	List registered skills
Server & Diagnostics	
<code>/webapi enable</code>	Enable HTTP REST API
<code>/webapi disable</code>	Disable HTTP REST API
<code>/doctor</code>	Run system diagnostics
<code>/update</code>	Check for updates
<code>/update install</code>	Install updates
UI & History	
<code>/log</code>	Toggle log panel visibility
<code>/history</code>	Browse previous inputs
<code>/inputdiag</code>	Input diagnostics
<code>/compact</code>	Compact context window

Command	Purpose
<code>/agent_compact</code>	Compact agent description

Benchmark Runner

The TUI exposes a native benchmark workflow through `/bench` using the currently selected provider/model and the shared `ragent-bench` crate.

- **Data roots:** `benches/data/<suite>/`
- **Result workbooks:** `benches/<suite>/<YYYY-MM-DD UTC>/<provider>/<model>.xlsx`
- **Virtual target:** `all` expands to every registered benchmark suite for both `/bench init` and `/bench run`
- **Virtual target:** `full` is reserved for complete upstream dataset ingestion across every suite and stays gated until all suites implement full-data initialization
- **Profiles:** `quick`, `standard`, and `agentic`
- **Init modes:** default `/bench init` writes local sample fixtures; `/bench init <suite> --full` performs full upstream dataset ingestion when the suite supports it
- **Background UX:** `/bench run ...` starts a background task, `/bench status` reports active or completed state, `/bench open last` prints the latest workbook path(s), and `/bench cancel` requests shutdown
- **Resume:** `--resume` reuses an existing same-day workbook only when benchmark, model, and config-hash sidecars match exactly

The benchmark workbook schema is fixed across suites (`run`, `metrics`, `cases`, `artifacts`) so HumanEval, MBPP, RepoBench, SWE-bench, and the native Phase 6 suites can be compared directly.

4.3 Key Bindings

Key	Action
<code>Enter</code>	Send message
<code>Ctrl+C</code>	Interrupt current operation
<code>Esc</code>	Clear input / Close overlay
<code>Tab</code>	Cycle focus between panels
<code>↑/↓</code>	Scroll message/log panels
<code>PgUp/PgDn</code>	Page scroll
<code>Home/End</code>	Jump to start/end
<code>Alt+V</code>	Paste image from clipboard
<code>Right-click</code>	Context menu (Cut/Copy/Paste)
<code>p</code>	Open provider setup
<code>? (empty input)</code>	Show keybindings help

4.4 TUI Component Architecture

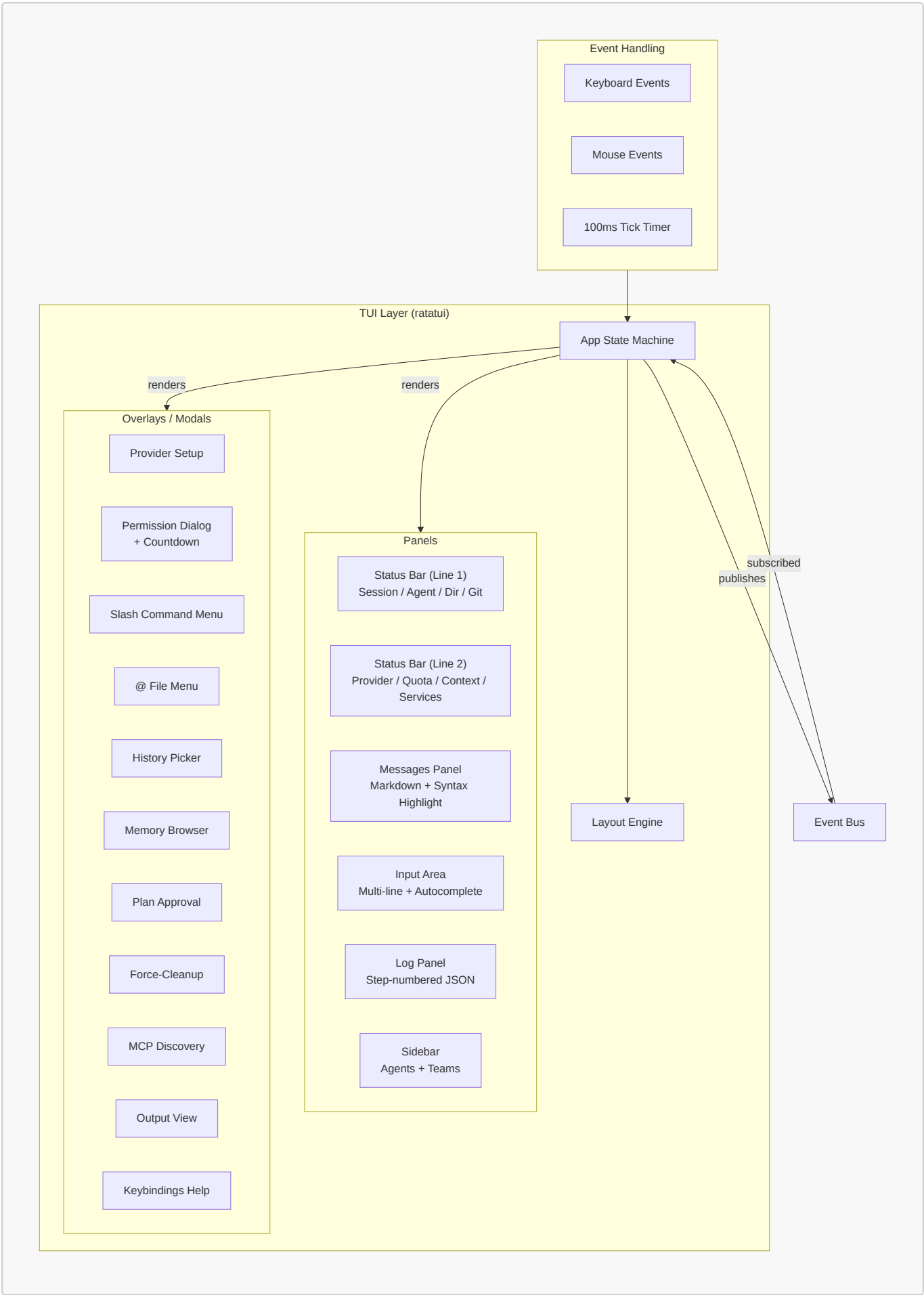


Figure 7: TUI Component Architecture — UI layout and event wiring

- **Streaming responses** — Real-time token streaming from LLM
- **Responsive two-line status bar** — Adapts between full, compact, and minimal layouts based on terminal width
- **Provider-aware usage display** — Shows quota percentage when available, otherwise token totals and context usage; Copilot plan labels and Ollama context labels are surfaced when known
- **Step-numbered tool calls** — Cross-session tool call correlation
- **Pretty-printed JSON** — Formatted tool parameters in log panel
- **Image attachments** — Visual support with clipboard paste
- **Mouse support** — Full mouse interaction
- **Auto-complete** — Slash command and agent name completion

5. HTTP Server & API

5.1 Server Commands

```
ragent serve                # Start server on default port (9100)
ragent serve --port 8080    # Custom port
```

5.2 API Endpoints

Health & Status

Method	Endpoint	Description
GET	/health	Health check - returns "ok"

Configuration & Providers

Method	Endpoint	Description
GET	/config	Get current application configuration
GET	/providers	List configured provider IDs

Sessions

Method	Endpoint	Description
GET	/sessions	List all sessions
POST	/sessions	Create new session
GET	/sessions/{id}	Get session details
DELETE	/sessions/{id}	Archive/delete a session
GET	/sessions/{id}/messages	Get messages for a session

Method	Endpoint	Description
POST	/sessions/{id}/messages	Send message (returns SSE stream)
POST	/sessions/{id}/abort	Abort an active session
POST	/sessions/{id}/permission/{req_id}	Reply to a permission request

Tasks (Background Agents)

Method	Endpoint	Description
GET	/sessions/{id}/tasks	List tasks for a session
POST	/sessions/{id}/tasks	Spawn a new background task
GET	/sessions/{id}/tasks/{tid}	Get task details
DELETE	/sessions/{id}/tasks/{tid}	Cancel a task

Server-Sent Events (SSE)

Method	Endpoint	Description
GET	/events	Global SSE event stream (all sessions)
GET	/sessions/{id}/messages	Session-specific SSE stream

Agents

Method	Endpoint	Description
GET	/agents	List available agents
GET	/agents/{name}	Get agent details

Tools

Method	Endpoint	Description
GET	/tools	List available tools
POST	/tools/{name}	Execute tool

Prompt Optimization

Method	Endpoint	Description
POST	/opt	Optimize prompt (requires Bearer token)

Memory API

Method	Endpoint	Description
GET	/memory/blocks	List memory blocks
GET	/memory/blocks/{scope}/{label}	Get specific block
PUT	/memory/blocks/{scope}/{label}	Create/update block
DELETE	/memory/blocks/{scope}/{label}	Delete block
POST	/memory/store	Store structured memory
POST	/memory/search	Search memories
GET	/memory/search	Search memories (query params)

Orchestrator API

Method	Endpoint	Description
POST	/orchestrate	Submit a job to the orchestrator
GET	/orchestrate/{job_id}	Get job status and results
DELETE	/orchestrate/{job_id}	Cancel a running orchestration job

Response Types

BlockResponse (Memory blocks):

```
{
  "scope": "project",
  "label": "conventions",
  "content": "Use snake_case...",
  "read_only": false,
  "created_at": "2025-01-15T10:30:00Z",
  "updated_at": "2025-01-15T12:00:00Z"
}
```

MemoryResponse (Structured memories):

```
{
  "id": "mem_abc123",
  "content": "The project uses PostgreSQL...",
  "category": "tech_stack",
  "confidence": 0.85,
  "tags": ["database", "infrastructure"],
  "created_at": "2025-01-15T10:30:00Z",
  "last_accessed": "2025-01-16T08:00:00Z"
}
```

Search Request Body (/memory/search):

```
{
  "query": "database configuration",
  "limit": 10,
  "semantic": true,
  "filters": {
    "category": "tech_stack",
    "min_confidence": 0.5
  }
}
```

5.3 Authentication

- Bearer token generated on server startup
- Token displayed in console: Server token: {token}
- Include in requests: Authorization: Bearer {token}

5.4 HTTP API Request Flow

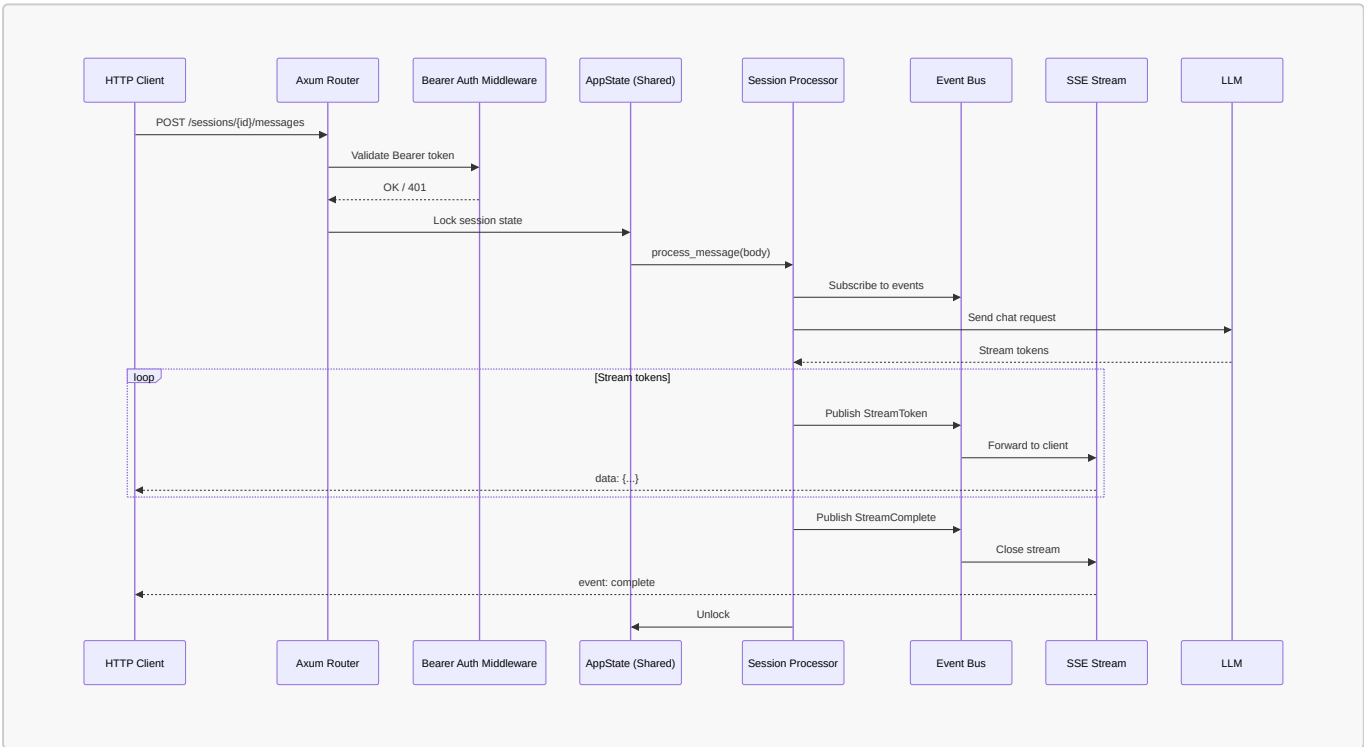


Figure 8: HTTP API Request Flow — REST + SSE lifecycle

Part II: Data & Knowledge Systems

6. Code Index

6.1 Overview

The Code Index is a built-in codebase indexing, search, and retrieval system that provides agents with deep, structured understanding of the codebase. Unlike simple text search (grep), it extracts symbols, their relationships, and enables semantic code exploration.

Key Features:

- **Zero external dependencies** — Everything compiles into the ragent binary (tree-sitter, SQLite, Tantivy)
- **User-controllable** — Enable/disable at any time via `/codeindex on|off`
- **Non-intrusive** — Zero overhead when disabled
- **Incremental updates** — Only re-indexes changed files using content hashing (Blake3)
- **Real-time file watching** — Automatic re-indexing on file changes
- **Fast search** — Sub-100ms symbol lookup across large codebases

6.2 Architecture

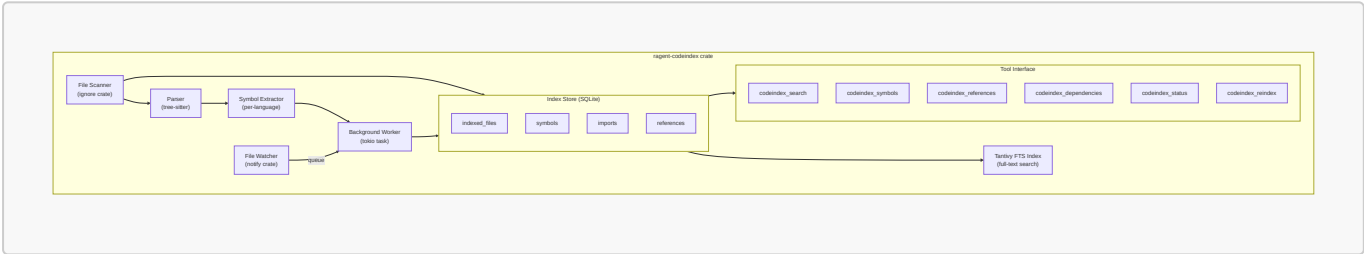


Figure 9: Code Index Pipeline — File scan → parse → index → search

Components:

Component	Purpose
File Scanner	Walk directory trees, respect <code>.gitignore</code> , compute content hashes
File Watcher	Real-time filesystem change detection via <code>notify</code> crate
Parser	Tree-sitter AST parsing with per-language grammar support
Symbol Extractor	Per-language AST walkers extract symbols, imports, and references
Index Store	SQLite persistence for files, symbols, imports, references
Search Engine	Tantivy full-text index + structured SQLite queries
Tree Cache	LRU cache of parse trees for incremental re-parsing
Background Worker	Async indexing worker with debounce, dedup, and batching

6.3 Supported Languages

Language	Extensions	Symbols Extracted
Rust	<code>.rs</code>	Functions, structs, enums, traits, impls, modules, consts, statics, type aliases, macros

Language	Extensions	Symbols Extracted
Python	.py	Functions, classes, methods, decorators, imports, async functions
TypeScript	.ts, .tsx	Functions, classes, interfaces, types, enums, namespaces, imports
JavaScript	.js, .jsx	Functions, classes, methods, arrow functions, imports
Go	.go	Functions, structs, interfaces, methods, imports, type definitions
C/C++	.c, .cpp, .h, .hpp	Functions, structs, unions, enums, classes, namespaces, includes
Java	.java	Classes, interfaces, enums, methods, constructors, annotations
OpenSCAD	.scad	Modules, functions, variable declarations, include/use statements, call references
Terraform	.tf, .tfvars	Resource blocks, data blocks, module calls, variables, locals, outputs, provider blocks
CMake	.cmake, CMakeLists.txt	Functions, macros, blocks, foreach/while loops, if conditions, commands, include/add_subdirectory
Gradle (Groovy)	.gradle	Classes, methods, functions, closures, imports, annotations, DSL block calls
Gradle (Kotlin)	.gradle.kts	Classes, functions, properties, type aliases, imports, companion objects, DSL calls
Maven	pom.xml	Project coordinates, dependencies, modules, plugins, profiles, properties, repositories

6.4 Data Model

Indexed Files

```

struct FileEntry {
    path: String,           // Relative path from project root
    language: String,       // "rust", "python", "typescript", etc.
    content_hash: String,   // Blake3 hash for change detection
    indexed_at: String,     // ISO 8601 timestamp
    file_size: i64,        // Bytes
}

```

Symbols

```
struct Symbol {
  name: String,           // Symbol name
  kind: SymbolKind,       // Function, Struct, Enum, Trait, etc.
  visibility: Visibility, // Public, Private, Restricted
  file_path: String,      // Source file path
  start_line: u32,        // 1-based line number
  start_col: u32,         // 1-based column
  end_line: u32,          // End line
  end_col: u32,           // End column
  doc: Option<String>,    // Doc comment / documentation
}
```

SymbolKind Taxonomy:

Kind	Description
<code>function</code>	Named function or method
<code>struct</code>	Struct or class definition
<code>enum</code>	Enum type
<code>trait</code>	Trait or interface definition
<code>impl</code>	Implementation block
<code>const</code>	Constant definition
<code>static</code>	Static variable
<code>type_alias</code>	Type alias
<code>module</code>	Module or namespace
<code>macro</code>	Macro definition
<code>field</code>	Struct/class field
<code>variant</code>	Enum variant

6.6 Control

```
/codeindex on           # Enable indexing
/codeindex off          # Disable indexing
/codeindex status       # Show current status
/codeindex reindex      # Force full re-index
/codeindex clear        # Delete all indexed data
```

Configuration in `ragent.json`:

```
{
  "code_index": {
    "enabled": true,
    "index_dir": ".ragent/code_index", // Custom location
    "max_file_size": 1048576, // 1MB default
    "extra_exclude_dirs": ["vendor", "node_modules", "target"],
    "extra_exclude_patterns": ["*.min.js", "*.d.ts"]
  }
}
```

6.7 Code Index Tools

All tools are available to agents and can be called directly in conversations. Because they perform local, read-only analysis, the codeindex tool family is hardwired as always allowed and bypasses interactive permission prompts.

codeindex_search

Full-text search across symbols, documentation, and code.

Parameters:

Parameter	Type	Description
query	string	Search query (supports boolean operators)
language	string?	Filter by language (e.g., "rust")
file_pattern	string?	Filter by file path pattern (e.g., "src/**/*.rs")
max_results	integer?	Maximum results (default: 20, max: 100)

Example:

```
{
  "query": "config parser",
  "language": "rust",
  "file_pattern": "crates/ragent-agent/**/*.rs",
  "max_results": 10
}
```

Returns: List of search results with symbol info, file path, and relevance score.

codeindex_symbols

Query symbols from the codebase index with optional filters.

Parameters:

Parameter	Type	Description
name	string?	Filter by symbol name (substring match)
kind	string?	Filter by symbol kind ("function", "struct", "enum", etc.)
file_path	string?	Filter by file path substring
language	string?	Filter by programming language
visibility	string?	Filter by visibility ("public", "private", "restricted")
limit	integer?	Maximum results (default: 50, max: 200)

Example:

```
{
  "name": "parse",
  "kind": "function",
  "language": "rust",
  "limit": 20
}
```

Returns: Structured symbol information with signatures and documentation.

codeindex_references

Find all references to a symbol by name across the indexed codebase.

Parameters:

Parameter	Type	Description
symbol	string	The symbol name to find references for
limit	integer?	Maximum results (default: 50, max: 200)

Example:

```
{
  "symbol": "AgentConfig",
  "limit": 100
}
```

Returns: File locations grouped by file, with reference kind (call, type, field_access).

codeindex_dependencies

Query file-level dependencies from the code index.

Parameters:

Parameter	Type	Description
<code>path</code>	string	File path to query dependencies for
<code>direction</code>	string?	"imports" (what this file uses) or "dependents" (what uses this file)

Example:

```
{
  "path": "crates/ragent-agent/src/agent/mod.rs",
  "direction": "dependents"
}
```

Returns: List of file paths that depend on (or are imported by) the target file.

`codeindex_status`

Show current status and statistics of the codebase index.

No parameters.

Returns:

- Files indexed
- Symbols extracted
- Languages detected
- Index size on disk
- Timestamps

Example Output:

```
{
  "files_indexed": 128,
  "symbols_extracted": 3427,
  "languages": {
    "rust": 89,
    "python": 23,
    "typescript": 16
  },
  "index_size_bytes": 2457600,
  "last_updated": "2026-04-14T09:30:00Z"
}
```

codeindex_reindex

Trigger a full re-index of the codebase. Use after major file changes or when search results seem stale.

No parameters.

Note: This can take several minutes for large codebases. Progress is shown in the TUI.

7. Memory System

The memory system provides persistent, structured storage of facts, patterns, preferences, and insights across sessions. It operates across three tiers with automatic extraction, decay, compaction, and optional semantic search.

7.1 Three-Tier Architecture

Tier	Storage	Access	Purpose
File Blocks	<code>~/.ragent/memory/</code> and <code>.ragent/memory/</code>	<code>memory_read</code> / <code>memory_write</code>	Human-readable markdown notes organised by topic
Structured Store	SQLite (<code>memory</code> table)	<code>memory_store</code> / <code>memory_recall</code> / <code>memory_forget</code>	Typed, tagged, confidence-scored facts with full-text search
Semantic Search	ONNX embeddings (optional)	<code>memory_search</code>	Embedding-based similarity search via <code>all-MiniLM-L6-v2</code>

7.2 Memory Tools

Tool	Purpose
<code>memory_read</code>	Read a named memory block (e.g. <code>memory_read(label="patterns")</code>)
<code>memory_write</code>	Write or append to a memory block with optional YAML frontmatter
<code>memory_replace</code>	Replace a specific string within a named memory block
<code>memory_store</code>	Store a structured memory with category, tags, and confidence score
<code>memory_recall</code>	Full-text search across structured memories with filtering
<code>memory_forget</code>	Delete memories by ID, age, confidence, category, or tags
<code>memory_search</code>	Semantic similarity search (embeddings-based)
<code>memory_migrate</code>	Analyse a flat <code>MEMORY.md</code> and propose splitting into named blocks

7.3 Memory Browser

The TUI provides a full-panel memory browser accessible via `/memory`:

- Lists global and project memory blocks
- Shows size indicators with warnings for blocks near the 64 KB limit
- Expand/collapse to view full content
- Keyboard navigation (**j/k**, **Enter**, **Esc**)
- Search and filter capabilities

7.4 Structured Memory Categories

Category	Description	Example
fact	Objective information about the project	"Uses tokio for async runtime"
pattern	Recurring code or workflow patterns	"Prefer anyhow::Result in main"
preference	User working preferences	"Use 4-space indentation"
insight	Deeper understanding or analysis	"Codebase follows clean architecture"
error	Known issues and their resolutions	"Don't use git checkout to rewind files"
workflow	Standard operating procedures	"Update CHANGELOG.md before pushing"

7.5 Auto-Extraction

When `memory_extraction_enabled` is true in the configuration, the agent automatically extracts memories from conversations. Key facts, patterns, and insights are identified and stored with appropriate categories and confidence scores.

7.6 HTTP API Endpoints

Method	Endpoint	Purpose
GET	/memory/blocks	List all memory blocks
GET	/memory/blocks/{label}	Read a specific memory block
DELETE	/memory/blocks/{label}	Delete a memory block
POST	/memory/search	Semantic search across memories
POST	/memory/store	Store a structured memory
POST	/memory/forget	Delete memories matching criteria

8. Spec Management

8.1 Overview

The Spec Management system provides a structured workflow for writing, tracking, and enforcing software specifications alongside code. It ensures that every significant feature or change is preceded by a clear, reviewable specification that lives in version control under the `specs/` directory.

Key Features:

- **Directory conventions** — Standardised `specs/<spec-id>/` layout with `SPEC.md` and `PLAN.md`
- **EARS templates** — Boilerplate generation with numbered requirement placeholders
- **Validation** — Automated checks for spec completeness and EARS syntax compliance
- **Status tracking** — State machine for spec lifecycle (draft → review → approved → implemented → verified → archived)
- **Plan linkage** — Mandatory `PLAN.md` pairing with every `SPEC.md`
- **Tool integration** — Slash commands and programmatic APIs for spec operations
- **Reporting** — Listing, filtering, and coverage summaries across a project

8.2 Directory Structure

```
specs/  
├── <spec-id>/  
│   ├── SPEC.md           # EARS-formatted specification  
│   └── PLAN.md           # Implementation plan with linked tasks  
├── <another-spec>/  
│   ├── SPEC.md  
│   └── PLAN.md  
└── ...
```

Rules enforced by the system:

- Every spec lives in a subdirectory of `specs/` named after the spec identifier
- Each spec directory must contain both `SPEC.md` and `PLAN.md`
- Spec IDs use only alphanumeric characters, hyphens, and underscores

8.3 Spec Lifecycle Status

Status	Meaning	Allowed Next States
draft	Initial creation, requirements being written	in_review
in_review	Under peer review	draft (with feedback), approved
approved	Review passed, ready for implementation	in_progress
in_progress	Being implemented	implemented
implemented	Code complete, awaiting verification	verified
verified	Tests pass, acceptance criteria met	archived
archived	Retired spec	draft (reopen)

8.4 Slash Commands

Command	Purpose
<code>/spec create <id> <feature></code>	Generate <code>SPEC.md</code> + <code>PLAN.md</code> via explore agent
<code>/spec list [status]</code>	List specs with optional status filter

Command	Purpose
<code>/spec search <query></code>	Full-text search across <code>SPEC.md</code> , <code>PLAN.md</code> , and <code>REVIEW.md</code>
<code>/spec validate [spec-id]</code>	Validate EARS compliance; all specs if no ID
<code>/spec status <id> <status></code>	Transition a spec to a new lifecycle status
<code>/spec task <id></code>	List tasks for a spec
<code>/spec activate <id></code>	Activate a spec for context injection into agent prompts
<code>/spec deactivate</code>	Deactivate the active spec
<code>/spec coverage <id></code>	Show requirement coverage report with linked tasks

8.5 Programmatic Tools

Five spec tools are available to agents for programmatic access:

Tool	Purpose
<code>spec_list</code>	List all specs with optional status filter
<code>spec_read</code>	Read a spec by ID — returns full markdown, requirements, tasks, and metadata
<code>spec_search</code>	Full-text search across spec content
<code>spec_coverage</code>	Generate a requirement coverage report
<code>spec_task_update</code>	Update the status of a task within a spec

8.6 EARS Notation

Requirements are written in EARS (Easy Approach to Requirements Syntax) notation. The system validates that every requirement conforms to one of five templates:

- 1. **Ubiquitous** — `The <system> shall <requirement>.`
- 2. **Event-Driven** — `When <event>, the <system> shall <requirement>.`
- 3. **State-Driven** — `While <state>, the <system> shall <requirement>.`
- 4. **Optional** — `Where <feature> is <configured>, the <system> shall <requirement>.`
- 5. **Unwanted** — `If <condition>, the <system> shall <requirement>.`

8.7 Active Spec Context Injection

When a spec is activated via `/spec activate <id>`:

- The spec's requirements and tasks are injected into the agent's system prompt
- The agent receives structured context about the spec's goals and acceptance criteria
- This keeps the agent aligned with the specification during implementation

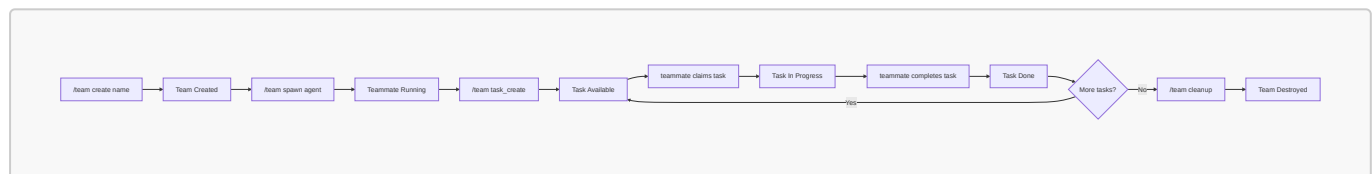
9. Teams

The team system enables multi-agent coordination with named teammates, shared task lists, mailbox messaging, and task assignment. Teams are persisted on disk and can be created, opened, closed, and cleaned up through slash commands.

9.1 Core Concepts

Concept	Description
Team	A named group of agents with shared state and tasks
Teammate	An individual agent instance within a team
Task	A work item with title, description, status, and optional dependencies
Mailbox	Message queue per teammate for asynchronous communication
Memory Scope	Per-teammate persistent memory (<code>user</code> , <code>project</code> , or <code>none</code>)

9.2 Team Lifecycle



9.3 Team Tools (21)

Tool	Purpose
<code>team_create</code>	Create a new team with a unique name
<code>team_spawn</code>	Spawn a teammate agent into the team
<code>team_cleanup</code>	Tear down a team and remove its on-disk resources
<code>team_status</code>	Get team member list, states, and task progress summary
<code>team_idle</code>	Signal that a teammate has no more tasks to work on
<code>team_task_create</code>	Add a new task to the team's shared task list
<code>team_task_claim</code>	Claim the next available task (or a specific task ID)
<code>team_task_complete</code>	Mark a claimed task as completed
<code>team_task_list</code>	List all tasks with status, assignment, and dependencies
<code>team_assign_task</code>	Assign a specific pending task to a named teammate
<code>team_message</code>	Send a direct message to one team member
<code>team_broadcast</code>	Send a message to all active teammates simultaneously
<code>team_read_messages</code>	Read unread messages from the teammate's mailbox
<code>team_shutdown_teammate</code>	Request graceful shutdown of a teammate (lead-only)

Tool	Purpose
<code>team_shutdown_ack</code>	Acknowledge a shutdown request and terminate
<code>team_submit_plan</code>	Submit a plan to the team lead for approval
<code>team_approve_plan</code>	Approve or reject a teammate's submitted plan (lead-only)
<code>team_wait</code>	Block until teammates finish their current work
<code>team_memory_read</code>	Read from a teammate's persistent memory directory
<code>team_memory_write</code>	Write to a teammate's persistent memory directory

9.4 Team Slash Commands

Command	Purpose
<code>/team create <name></code>	Create new team
<code>/team open <name></code>	Open existing team
<code>/team close</code>	Close team session
<code>/team delete <name></code>	Delete team
<code>/team clear</code>	Clear team state
<code>/team tasks</code>	Show team tasks table
<code>/team status</code>	Show team status
<code>/team message <to> <content></code>	Send message to teammate
<code>/team broadcast <content></code>	Broadcast to all teammates
<code>/team spawn <agent></code>	Spawn teammate agent
<code>/team cleanup</code>	Cleanup team resources

9.5 Task Dependencies

Tasks can declare dependencies on other tasks. A task cannot be claimed until all its dependencies are completed. This enables sequential workflows within parallel teams.

9.6 Plan Approval Workflow

Teammates can submit plans to the team lead for approval before starting implementation:

1. Teammate calls `team_submit_plan` with their intended approach
2. Lead receives the plan via `team_read_messages`
3. Lead calls `team_approve_plan` with `approved: true` or `approved: false`
4. If approved, teammate exits plan-pending mode and begins implementation
5. If rejected, teammate receives feedback and revises

10. Swarm Mode

Swarm mode automatically decomposes a high-level goal into parallel subtasks, creates an ephemeral team, and coordinates execution. It is ideal for large tasks that can be broken into independent work items.

10.1 How Swarm Works

1. **Decomposition** — The LLM analyses the prompt and produces a JSON decomposition with 2–8 independent subtasks and optional dependency edges.
2. **Team Creation** — An ephemeral team named `swarm-<timestamp>` is created.
3. **Teammate Spawning** — One teammate is spawned per subtask with a tailored prompt.
4. **Dependency Resolution** — Tasks with `depends_on` are blocked until prerequisites complete.
5. **Progress Monitoring** — The TUI shows real-time status: spawning, blocked, in-progress, and completed counts.
6. **Auto-Completion** — When all tasks finish, the swarm auto-summarises results.

10.2 Swarm Slash Commands

Command	Purpose
<code>/swarm <prompt></code>	Decompose a goal into parallel subtasks and spawn a team
<code>/swarm status</code>	Show live progress of the active swarm
<code>/swarm cancel</code>	Cancel the active swarm and clean up
<code>/swarm help</code>	Show usage help

10.3 Decomposition Format

The LLM returns a JSON object with an array of subtasks:

```
{
  "tasks": [
    {
      "id": "s1",
      "title": "Short title",
      "description": "Detailed instructions for the agent...",
      "depends_on": [],
      "agent_type": "general",
      "model": null
    }
  ]
}
```

Each subtask has:

- `id` — Unique identifier (e.g. `s1`, `s2`)
- `title` — Short human-readable title
- `description` — Full instructions for the teammate

- `depends_on` — IDs of subtasks that must complete first
- `agent_type` — Optional agent type override (defaults to `general`)
- `model` — Optional model override in `provider/model` format

10.4 Swarm State

The TUI tracks swarm state including:

- `team_name` — Name of the ephemeral backing team
- `prompt` — Original user prompt
- `decomposition` — The LLM-produced task breakdown
- `spawned` — Whether all teammates have been spawned
- `completed` — Whether the orchestrator has collected all results

10.5 Limitations

- Only one swarm can be active at a time; start a new one after cancelling the current swarm.
- Requires a configured model with JSON mode support for reliable decomposition.
- Subtask descriptions must be detailed enough for agents to work without further clarification.

11. Autopilot Mode

Autopilot mode enables the agent to operate autonomously, continuing to iterate on a task without user intervention until the task is complete, a safety limit is reached, or the user stops it.

11.1 How Autopilot Works

1. **Activation** — User runs `/autopilot on` with optional token and time budgets.
2. **Autonomous Loop** — The agent processes messages, makes tool calls, and continues iterating automatically.
3. **Completion** — The agent calls `task_complete` with a summary, or the user runs `/autopilot off`.
4. **Safety Limits** — Hard stops prevent runaway execution.

11.2 Autopilot Slash Commands

Command	Purpose
<code>/autopilot on [--max-tokens N] [--max-time N]</code>	Enable autonomous operation with optional limits
<code>/autopilot off</code>	Disable autonomous operation and return to interactive mode
<code>/autopilot status</code>	Show autopilot status, elapsed time, and remaining budget

11.3 Safety Limits

Limit	Default	Behaviour When Hit
max_steps	500	Halt and ask user whether to continue
Token budget	Optional (--max-tokens)	Stop and display summary
Time limit	Optional (--max-time in seconds)	Stop and display summary
Context window	Model-specific	Trigger automatic compaction

11.4 Completion Signalling

Agents in autopilot mode call `task_complete` to signal completion:

```
task_complete(summary: "Implemented feature X with tests and documentation")
```

This publishes a `TaskCompleted` event, displays the summary to the user, and exits autopilot mode.

11.5 Status Display

When active, the status bar shows:

- ⚡ `autopilot` — Normal operation
- `autopilot: time limit reached` — Time budget exhausted
- `autopilot stopped: task complete` — Agent finished successfully

12. Orchestrator & Multi-Agent Coordination

The orchestrator provides primitives for coordinating multiple agents in a single workflow. It supports job dispatch, progress tracking, and result aggregation.

12.1 Core Components

Component	Purpose
AgentRegistry	Maintains a registry of available agents and their capabilities
Coordinator	Dispatches jobs to agents and collects results
InProcessRouter	Actor-style message routing between agents
JobDescriptor	Defines a job with required capabilities and payload

12.2 Coordinator API

```
use ragent_agent::orchestrator::{Coordinator, JobDescriptor};

let coord = Coordinator::new(registry);
```

```
// Synchronous job (blocks until all agents respond)
let result = coord.start_job_sync(JobDescriptor {
  id: "job-1".to_string(),
  required_capabilities: vec!["search".to_string()],
  payload: "find TODOs".to_string(),
}).await?;

// Asynchronous job (returns job ID, subscribe to events)
let job_id = coord.start_job_async(desc).await?;
let mut events = coord.subscribe_job_events(&job_id).await?;
```

12.3 HTTP API Endpoints

Method	Endpoint	Purpose
GET	/orchestrator/metrics	Return live counter snapshot
POST	/orchestrator/start	Start a multi-agent job
GET	/orchestrator/jobs/{id}	Poll job status and results

12.4 Conflict Resolution

The **policy** submodule provides conflict resolution strategies:

- **Last-write-wins** — Simple timestamp-based resolution
- **Human-in-the-loop** — Escalate conflicts to the user for decision
- **Merge strategies** — Automatic merging for compatible changes

12.5 Transport Adapters

The **transport** submodule supports pluggable communication:

- **InProcessRouter** — In-process actor-style messaging
- **HttpRouter** — HTTP-based inter-process communication
- **RouterComposite** — Combine multiple transport layers

12.6 Current Status

The orchestrator is at MVP level with in-process coordination. HTTP endpoints are available but the full distributed coordination (leader election, cluster formation) is planned for a future milestone.

13. Custom Agents

Custom agents extend ragent's built-in agent personalities with user-defined profiles. Agents are stored as JSON (OASF format) or Markdown files and loaded automatically at startup.

13.1 Discovery Paths

Custom agents are discovered from two locations, with project-local taking precedence:

Priority	Directory
1 (lower)	<code>~/.ragent/agents/</code>
2 (higher)	<code>[PROJECT]/.ragent/agents/</code>

13.2 File Formats

OASF JSON Format

```
{
  "name": "my-reviewer",
  "description": "Code reviewer focused on security",
  "version": "1.0.0",
  "schema_version": "0.7.0",
  "modules": [{
    "type": "ragent/agent/v1",
    "payload": {
      "system_prompt": "You are an expert code reviewer...",
      "mode": "primary",
      "max_steps": 30,
      "thinking": { "enabled": true, "level": "high" },
      "permissions": [
        { "permission": "file:write", "pattern": "src/**", "action":
"allow" }
      ],
      "skills": ["security-review", "rust-linting"]
    }
  }]
}
```

Markdown Format

Uses YAML frontmatter between `---` delimiters with the markdown body as the system prompt:

```
---
name: my-reviewer
version: 1.0.0
model: anthropic/claude-sonnet-4-20250514
skills:
  - security-review
permissions:
  - permission: file:write
    pattern: src/**
    action: allow
---

You are an expert code reviewer focused on security vulnerabilities...
```

13.3 Agent Configuration Fields

Field	Type	Description
<code>system_prompt</code>	string	The agent's personality and instructions
<code>mode</code>	string	<code>primary</code> (main agent) or <code>secondary</code> (helper)
<code>max_steps</code>	integer	Maximum tool calls per turn
<code>thinking</code>	object	Reasoning configuration: { <code>enabled</code> , <code>level</code> , <code>budget_tokens</code> }
<code>permissions</code>	array	Per-agent permission rules
<code>skills</code>	array	Skills this agent can invoke
<code>model</code>	string	Default model in <code>provider/model</code> format
<code>memory_scope</code>	string	<code>user</code> , <code>project</code> , or <code>none</code>

13.4 Agent Diagnostics

Use `/agents` to list all loaded agents including custom ones. Custom agents are marked with a yellow `[custom]` badge. Diagnostics show any load errors or skipped files.

13.5 OASF Annotations

OASF records support taxonomy annotations for discoverability:

- `skills` — OASF skill taxonomy (e.g. `software_engineering/code_review`)
- `domains` — OASF domain taxonomy (e.g. `technology/software_development`)
- `locators` — Source code or registry references

14. Skills System

Skills are loadable instruction packs that inject tools, prompts, and file context into agent sessions. Each skill is defined by a `SKILL.md` file with YAML frontmatter and a markdown body.

14.1 Skill Discovery

Skills are discovered from multiple sources with priority order:

Priority	Scope	Path
0	Bundled	Built into ragent binary
1	Enterprise	Managed settings
2	OpenSkills Global	<code>~/.agent/skills/</code> , <code>~/.claude/skills/</code>
3	Personal	<code>~/.ragent/skills/<name>/SKILL.md</code>
4	OpenSkills Project	<code>.agent/skills/</code> , <code>.claude/skills/</code>
5	Project	<code>.ragent/skills/<name>/SKILL.md</code>

Higher-priority scopes override lower ones when names conflict.

14.2 Skill Structure

```
.ragent/skills/
  deploy/
    SKILL.md          # Skill instructions and frontmatter (required)
    scripts/          # Helper scripts the skill can invoke
    templates/        # Template files for the agent to fill in
    examples/         # Example outputs showing expected format
    resources/        # Reference materials
```

14.3 SKILL.md Frontmatter

```
---
name: deploy
version: 1.0.0
description: Deploy the application to staging
argument_hint: "[environment]"
user_invocable: true
disable_model_invocation: false
allowed_tools:
  - bash
  - write
model: "anthropic/claude-sonnet-4-20250514"
context: fork
agent: general-purpose
---

# Deploy Skill

When the user types `/deploy [environment]`, deploy the application...
```

14.4 Skill Fields

Field	Description
name	Unique identifier (lowercase, hyphens, max 64 chars)
description	What the skill does; used for auto-invocation matching
argument_hint	Shown during autocomplete (e.g. "[environment]")
user_invocable	If false, hidden from / menu; only agent can invoke
disable_model_invocation	If true, only user can invoke via /name
allowed_tools	Tools the agent can use without permission when this skill is active
model	Override model when this skill is active

Field	Description
<code>context</code>	<code>fork</code> to run in a forked subagent context
<code>agent</code>	Subagent type when <code>context</code> is <code>fork</code>

14.5 Invocation

Skills are invoked by including `/skillname` in a message or by the agent auto-invoking based on description matching. Arguments after the skill name are passed to the skill body via template substitution.

14.6 Template Variables

Skill bodies support variable substitution:

Variable	Description
<code>\$0</code>	Full argument string
<code>\$1, \$2, ...</code>	Individual arguments
<code>\${RAGENT_SKILL_DIR}</code>	Directory containing the skill's <code>SKILL.md</code>
<code>\${RAGENT_SESSION_ID}</code>	Current session ID
<code>\${RAGENT_WORKING_DIR}</code>	Current working directory

14.7 Bundled Skills

Ragent ships with 4 bundled skills:

Skill	Purpose
<code>simplify</code>	Review recently changed files for code quality, reuse, and efficiency issues
<code>debug</code>	Troubleshoot current session by reading debug logs
<code>security-audit</code>	Security-focused code review
<code>test-coverage</code>	Analyse test coverage gaps

14.8 Dynamic Context Injection

Skills can enable dynamic context with `allow_dynamic_context: true`. This allows `!command` syntax within the skill body to execute shell commands and inject their output into the context.

15. Prompt Optimization

The prompt optimization system transforms plain prompts into structured frameworks using template-based meta-prompts. No external API calls are needed — the optimization is performed by sending the framework's system prompt to the current LLM provider.

15.1 Optimization Methods

Method	Name	Description
<code>co_star</code>	CO-STAR	Context, Objective, Identity, Tone, Audience, Result
<code>crispe</code>	CRISPE	Capacity/Role, Request, Intent, Steps, Persona, Examples
<code>cot</code>	Chain-of-Thought	Step-by-step reasoning scaffold with self-checks
<code>draw</code>	DRAW	Professional AI image prompt optimizer
<code>rise</code>	RISE	Recursive Introspection — iterative self-improvement loop
<code>o1_style</code>	O1-STYLE	Thinking/step/reflection/reward structured reasoning
<code>meta</code>	Meta Prompting	Distil to a concise, high-signal meta-prompt
<code>variational</code>	VARI	Variational planning content-generation scaffold
<code>q_star</code>	Q*	XML Q*/A* intelligent iterative prompt optimizer
<code>openai</code>	OpenAI	Detailed GPT-style system prompt with guidelines
<code>claude</code>	Claude	Anthropic-style XML instruction generator with examples
<code>microsoft</code>	Microsoft	Azure AI optimised prompt with quality targets

15.2 Usage

TUI:

```
/opt help           # Show method table
/opt co_star Explain lifetimes
/opt cot Solve two-sum
/opt draw A futuristic city at sunset
```

HTTP API:

```
curl -s -X POST http://localhost:9100/opt \
  -H "Authorization: Bearer $RAGENT_TOKEN" \
  -H "Content-Type: application/json" \
  -d '{"method": "co_star", "prompt": "Explain Rust lifetimes"}'
```

15.3 How It Works

1. The user selects a method and provides a raw prompt.
2. The system loads the method's meta-prompt (system message) from the template library.
3. The meta-prompt and user's input are sent to the current LLM provider.
4. The LLM returns the optimised, structured prompt.
5. The result is displayed in the TUI or returned via the HTTP API.

15.4 Completer Trait

The optimization is decoupled from any specific LLM backend via the `Completer` trait:

```
#[async_trait]
pub trait Completer: Send + Sync {
    async fn complete(&self, system: &str, user: &str) ->
    anyhow::Result<String>;
}
```

Implementations connect to the session's active provider and model.

15.5 Method Aliases

Methods can be referenced by multiple aliases for convenience:

Method	Aliases
co_star	costar, co-star
crispe	crisper
cot	chain_of_thought, chain-of-thought, chainofthought
o1_style	o1-style, o1
meta	meta_prompting, meta-prompting
variational	vari
q_star	qstar, q-star, q*
microsoft	ms, azure

16. Configuration

16.1 Configuration Files

File	Purpose
ragent.json	Project-level configuration
ragent.jsonc	Project-level (with comments)
~/.config/ragent/config.json	User-global configuration

16.2 Configuration Schema

```
{
  "provider": {
    "anthropic": {
      "env": ["ANTHROPIC_API_KEY"],
      "thinking": { "enabled": true, "level": "low" },

```

```

    "models": {
      "claude-sonnet-4-20250514": {
        "thinking": { "enabled": true, "level": "high", "budget_tokens":
16000 }
      }
    },
    "openai": { /* ... */ },
    "copilot": { /* ... */ },
    "ollama": { /* ... */ },
    "generic_openai": { /* ... */ },
    "azure_foundry": {
      "env": ["AZURE_AI_FOUNDRY_API_KEY"],
      "api": { "base_url": "https://your-endpoint.azure.com" },
      "thinking": { "enabled": true, "level": "low" }
    }
  },
  "defaultAgent": "coder",
  "permissions": [],
  "skill_dirs": [],
  "code_index": {
    "enabled": true,
    "max_file_size": 1048576
  },
  "memory": {
    "auto_extract": { "enabled": false, "require_confirmation": true },
    "semantic": { "enabled": false, "dimensions": 384 },
    "compaction": { "enabled": true, "block_size_limit": 4096 },
    "eviction": { "auto": false, "stale_days": 30 }
  },
  "hidden_tools": ["github_list_issues", "gitlab_list_mrs"],
  "bash": {
    "allowlist": [],
    "denylist": []
  },
  "hooks": [
    { "trigger": "on_session_start", "command": "echo 'Session started'" }
  ]
}

```

Additional top-level configuration keys:

- **hidden_tools** — List of tool names to hide from LLM tool definitions and system-prompt tool listings. Hidden tools remain registered and executable; they are simply not advertised to the model. When configs are merged across layers, **hidden_tools** is unioned so entries from both global and project configs are honoured.
- **provider.<id>.thinking** — Provider-wide default reasoning configuration used when a selected model has no more specific override.
- **provider.<id>.models.<model>.thinking** — Per-model default reasoning configuration. Precedence is user selection → agent default → model config → provider config → built-in default.

Thinking configuration uses the shared **ThinkingConfig** type:

- `level` — `auto`, `off`, `low`, `medium`, or `high`
- `enabled` — explicit on/off switch for providers that separate enablement from effort
- `budget_tokens` — optional Anthropic thinking budget
- `display` — optional Anthropic display mode (`full`, `summarized`, `omitted`)

`ThinkingLevel` defaults to `auto`. In the TUI, `/thinking` changes the current session's effective reasoning level while model picker defaults still follow the precedence above.

16.3 Environment Variables

Variable	Purpose
<code>ANTHROPIC_API_KEY</code>	Anthropic API key
<code>OPENAI_API_KEY</code>	OpenAI API key
<code>GENERIC_OPENAI_API_KEY</code>	Generic OpenAI-compatible key
<code>GITHUB_COPILOT_TOKEN</code>	GitHub Copilot token
<code>OLLAMA_HOST</code>	Ollama server URL
<code>RAGENT_LOG_LEVEL</code>	Log level (trace/debug/info/warn/error)
<code>RAGENT_YES</code>	Auto-approve all permissions

16.4 Configuration Error Reporting

Configuration parsing errors are surfaced with actionable diagnostics rather than only a generic parse failure. When JSON or JSONC parsing fails, `ragent` reports:

- the full config file path,
- the line and column number,
- the problematic source line, and
- a caret (^) pointing at the error position.

This applies to normal config loading and explicit `--config` CLI usage, helping users fix malformed configuration files quickly.

Part V: External Integrations

17. GitLab Integration

`ragent` provides native GitLab integration through the `ragent-tools-vcs` crate, enabling agents to manage issues, merge requests, pipelines, and project metadata via the GitLab REST API.

17.1 Authentication

Configuration:

- **Instance URL:** The GitLab instance URL (e.g., <https://gitlab.com> or a self-hosted instance)
- **Personal Access Token (PAT):** Stored securely in the ragent SQLite database via `/gitlab setup`

Slash Commands:

Command	Purpose
<code>/gitlab setup</code>	Configure GitLab connection (instance URL + PAT)
<code>/gitlab logout</code>	Remove stored GitLab credentials
<code>/gitlab status</code>	Show GitLab connection status

17.2 Issue Tools

Tool	Description	Parameters
<code>gitlab_issues_list</code>	List issues with filtering	<code>project_id</code> , <code>state</code> (opened/closed/all), <code>labels</code> , <code>limit</code>
<code>gitlab_issues_get</code>	Get issue details	<code>project_id</code> , <code>issue_iid</code>
<code>gitlab_issues_create</code>	Create a new issue	<code>project_id</code> , <code>title</code> , <code>body</code> , <code>labels</code> , <code>assignee_ids</code>
<code>gitlab_issues_comment</code>	Add comment to an issue	<code>project_id</code> , <code>issue_iid</code> , <code>body</code>
<code>gitlab_issues_close</code>	Close an issue	<code>project_id</code> , <code>issue_iid</code>

17.3 Merge Request Tools

Tool	Description	Parameters
<code>gitlab_mr_list</code>	List merge requests	<code>project_id</code> , <code>state</code> , <code>target_branch</code> , <code>limit</code>
<code>gitlab_mr_get</code>	Get MR details and diff	<code>project_id</code> , <code>mr_iid</code>
<code>gitlab_mr_create</code>	Create a new merge request	<code>project_id</code> , <code>title</code> , <code>body</code> , <code>source_branch</code> , <code>target_branch</code>
<code>gitlab_mr_merge</code>	Merge a merge request	<code>project_id</code> , <code>mr_iid</code> , <code>squash</code>

17.4 CI/CD Pipeline Tools

Tool	Description	Parameters
<code>gitlab_pipeline_list</code>	List pipelines	<code>project_id</code> , <code>status</code> , <code>limit</code>
<code>gitlab_pipeline_get</code>	Get pipeline details	<code>project_id</code> , <code>pipeline_id</code>
<code>gitlab_ci_list</code>	List CI jobs for a pipeline	<code>project_id</code> , <code>pipeline_id</code>
<code>gitlab_ci_get</code>	Get CI job details and logs	<code>project_id</code> , <code>job_id</code>

17.5 Project Tools

Tool	Description	Parameters
gitlab_project_get	Get project metadata	project_id or path_with_namespace

17.6 Auto-Detection

When operating inside a Git repository with a GitLab remote, ragent can auto-detect the project_id or path_with_namespace from the git remote configuration:

```
git remote get-url origin
→ https://gitlab.com/owner/repo.git → path_with_namespace="owner/repo"
→ git@gitlab.com:owner/repo.git → path_with_namespace="owner/repo"
```

Falls back to explicit project_id or path_with_namespace parameters if detection fails.

Part VII: Security & Operations

24. Security & Permissions

24.1 Permission Security Layers

The permission system is a multi-layered defense-in-depth architecture that controls every tool invocation.

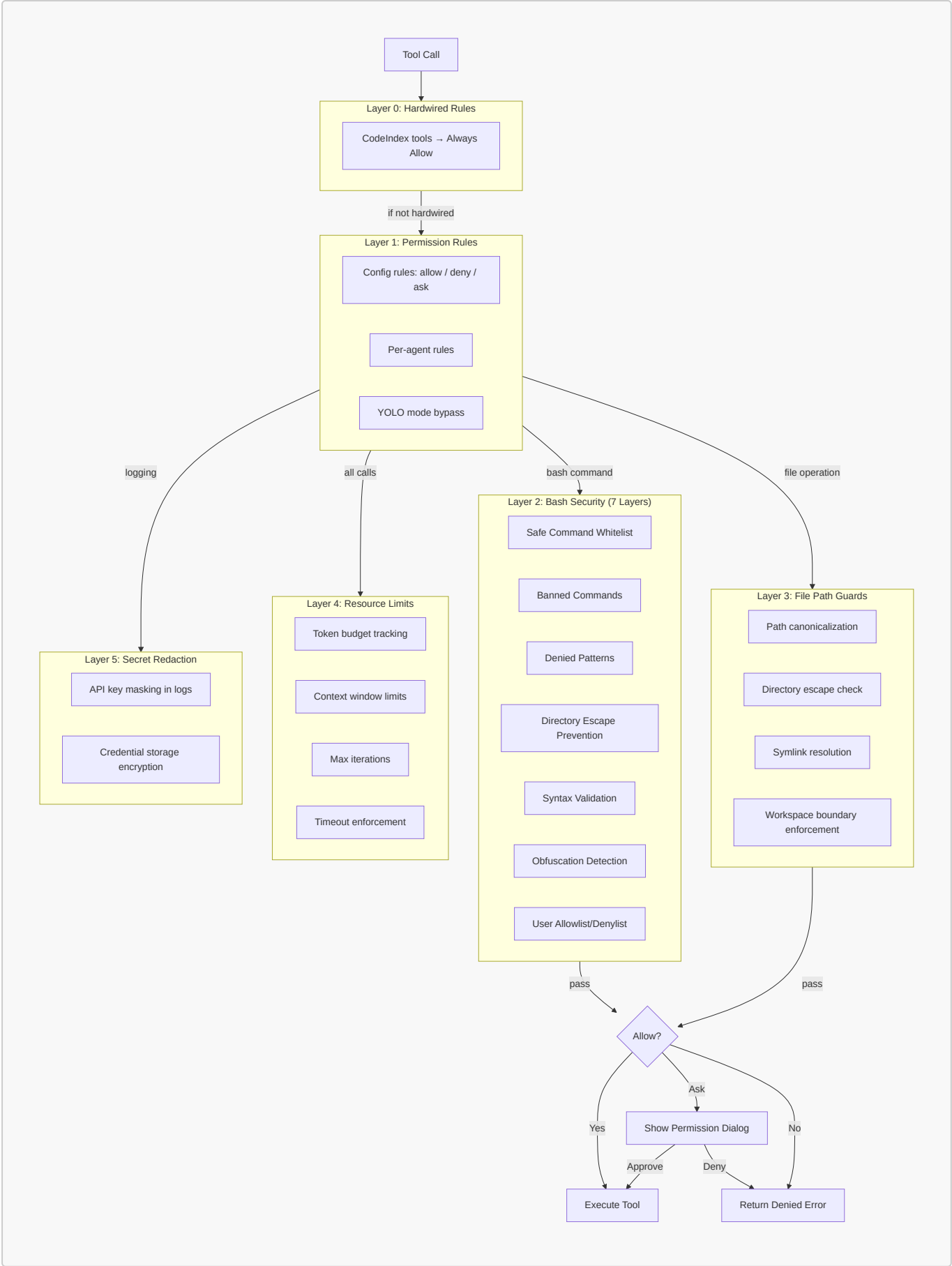


Figure 10: Permission Security Layers — 5-layer defense-in-depth

24.2 Bash Security — 7 Layers

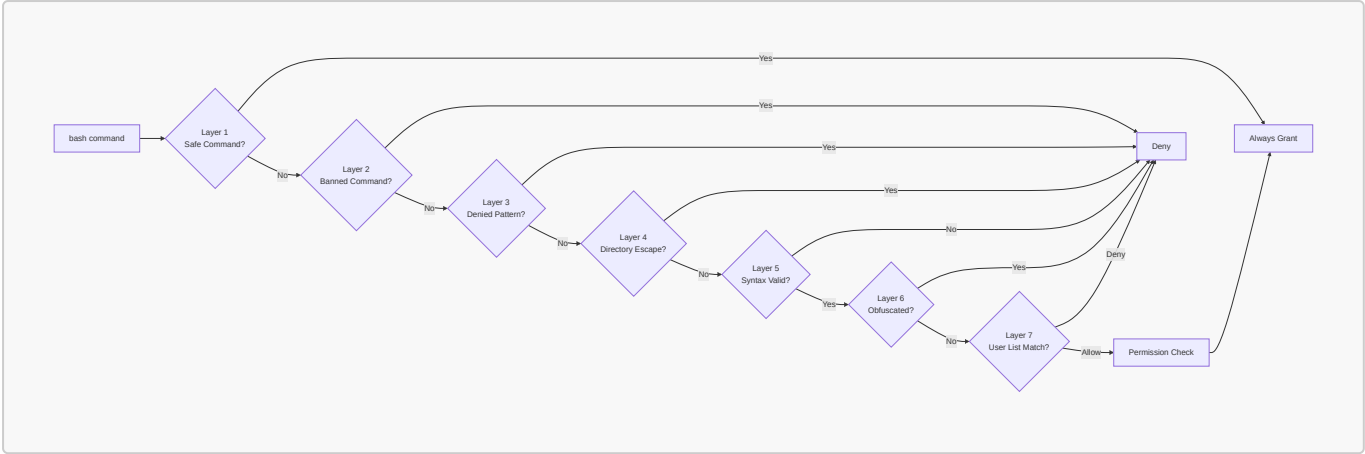


Figure 11: Bash Security — 7 Layers — Bash command defense flow

Layer Details:

Layer	Name	Description	Test Count
1	Safe Command Whitelist	51 commands auto-approved (cat, ls, git, cargo, etc.)	15
2	Banned Commands	22 commands always blocked (mkfs, fdisk, useradd, etc.)	6
3	Denied Patterns	46 destructive patterns (rm -rf /, fork bombs, etc.)	8
4	Directory Escape Prevention	Blocks cd/pushd outside workspace	4
5	Syntax Validation	Runs <code>sh -n -c</code> with 1s timeout	3
6	Obfuscation Detection	Detects <code>base64 bash</code> , <code>python exec</code> , hex escapes	5
7	User Allowlist/Denylist	User-configurable via <code>/bash allow/deny</code>	4

24.3 Permission Request Flow

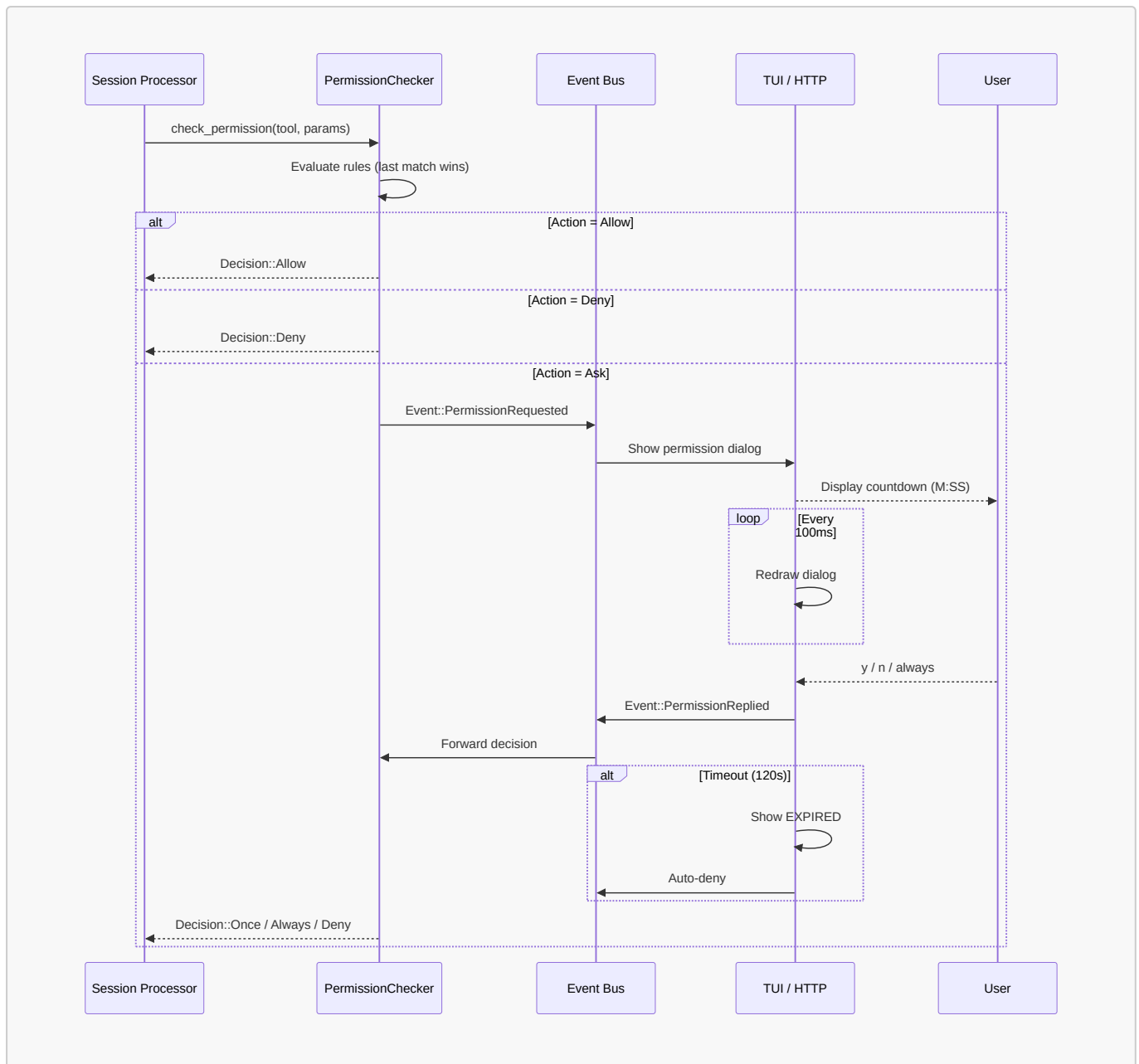


Figure 12: Permission Request Flow — From tool call to user decision

24.4 Permission Rules Evaluation

Rules are evaluated in order, with **last match wins** semantics:

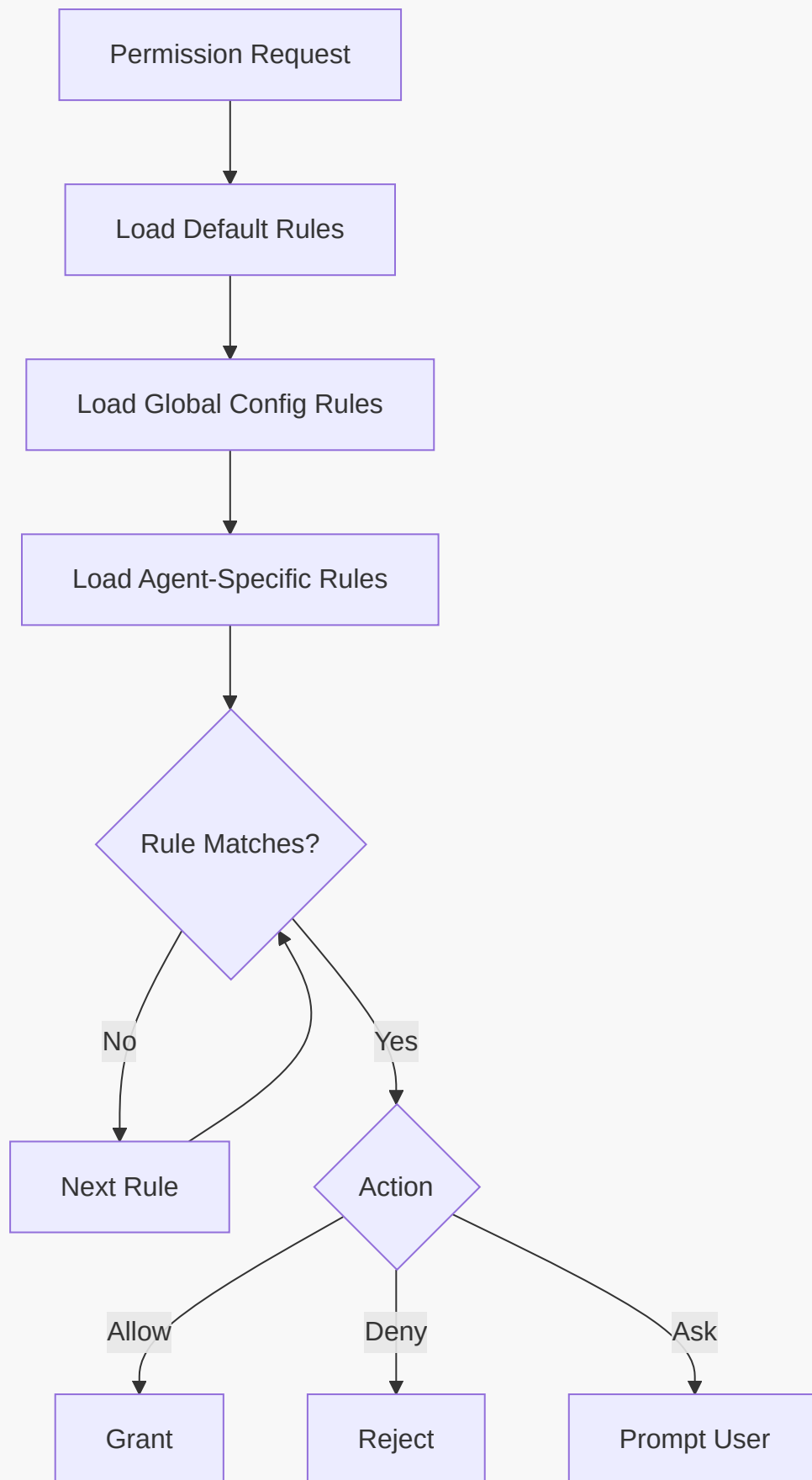


Figure 13: Permission Rules Evaluation — Rule matching and resolution

Default Rules:

- Read operations → Allow
- Edit operations → Ask
- Bash execution → Ask
- Web access → Ask
- Todo management → Allow

25. Auto-Update Mechanism

(To be documented)

Appendices

Appendix A: Version History

Version	Date	Highlights
v0.1.0-alpha.82	2026-05-20	Azure AI Foundry provider fixes, <code>/config show</code> slash command
v0.1.0-alpha.79	2026-05-18	Azure endpoint logging in TUI log panel
v0.1.0-alpha.76	2026-05-18	Azure AI Foundry provider added
v0.1.0-alpha.75	2026-05-18	SPEC.md mermaid diagram syntax fixes
v0.1.0-alpha.73	2026-05-18	<code>/model</code> selection fix, version number display
v0.1.0-alpha.72	2026-05-18	gen-spec-pdf.sh script, Spec Management section
v0.1.0-alpha.71	2026-05-18	Startup ASCII art banner with compile timestamp
v0.1.0-alpha.70	2026-05-17	Concurrency improvements, todo fixes
v0.1.0-alpha.68	2026-05-15	<code>/codeindex lang</code> filtering, benchmark data cleanup
v0.1.0-alpha.61	2025-01-17	Instruction file discovery logging
v0.1.0-alpha.60	2025-01-16	<code>.local/share/ragent</code> in AGENTS.md search path

Version	Date	Highlights
v0.1.0-alpha.57	2025-01-16	MS Office/LibreOffice fixes, todo summary updates
v0.1.0-alpha.49	2025-01-17	Permission dialog live countdown, config parse error enhancement, codeindex hardwired permissions, crate extraction milestones
v0.1.0-alpha.48	2025-01-17	Permission milestones complete, bash security layers, more permissions fixes
v0.1.0-alpha.47	2025-01-17	Crate reorganisation (ragent-types, ragent-config, ragent-storage, ragent-llm)

Appendix B: Documentation

All documentation markdown files are located in [docs/](#) except for these root files:

File	Purpose
README.md	Project overview
QUICKSTART.md	Quick start guide
SPEC.md	This specification
AGENTS.md	Agent guidelines
CHANGELOG.md	Change log
RELEASE.md	Release notes
STATS.md	Project statistics

Appendix C: Project Contact & Repository

- **Repository:** <https://github.com/thawkins/ragent>
- **License:** MIT
- **Author:** Tim Hawkins

Appendix D: Changelog (2025-01-16 → 2025-04-21)

Added (v0.1.0-alpha.76 → v0.1.0-alpha.82)

- Azure AI Foundry provider — New [azure_foundry](#) provider for Microsoft Azure AI Foundry models
- Azure endpoint logging — Full resolved endpoint URL displayed in TUI log panel
- [/config show](#) slash command — Displays current resolved configuration
- gen-spec-pdf.sh script — Pandoc + Chromium-based Markdown-to-PDF conversion
- Startup ASCII art banner — Application name in ASCII art with compile timestamp
- [/codeindex lang](#) filtering — Optional language parameter for code index results
- Instruction file discovery logging — Tracks AGENTS.md-style file discovery with summary

Changed (v0.1.0-alpha.68 → v0.1.0-alpha.75)

- Improved TUI slash-command autocomplete with safe `Esc` handling
- SPEC.md mermaid diagram syntax fixes — All 14 diagrams now render correctly
- Benchmark data cleanup — Removed unused dataset files

Fixed (v0.1.0-alpha.76 → v0.1.0-alpha.82)

- Azure endpoint logging now shows full URL with model name
- `/model` selection handling fixed

Earlier Changes (v0.1.0-alpha.47 → v0.1.0-alpha.68)

- Permission dialog countdown timer with live TUI updates (120-second timeout)
 - Config parse error reporting with file path, line, column, and caret marker
 - Codeindex tools hardwired as always-allowed (read-only, no permission prompts)
 - Crate extraction milestones: `ragent-types`, `ragent-config`, `ragent-storage`, `ragent-llm`
 - `ollama_cloud` provider with dynamic model discovery and vision support
 - `gemini` provider with massive context windows (up to 2M tokens)
 - `huggingface` provider with dynamic model discovery and rate limit tracking
 - GitLab integration with issues, merge requests, pipelines, and jobs
 - Team coordination tools (21 tools for team lifecycle, tasks, messaging)
 - MCP (Model Context Protocol) client support
 - Skills system for loadable skill packs
 - Custom agent profiles via OASF format
 - Autopilot mode for autonomous operation
 - Prompt optimization (`/opt` command with 12 methods)
 - Memory system with three tiers (file blocks, SQLite store, semantic search)
 - Journal system for insights and decisions
 - Background agent spawning and management
 - Swarm mode for parallel task decomposition
 - Plan agent with human-in-the-loop approval
 - Enhanced bash security with 7 layers and word-boundary matching
 - Permission system now supports per-agent rules and YOLO mode
-